FIS-2024-01560

Status: **Submitted**

Submission-Date: **18/03/2025, 10:30**

SECTION A - ADMINISTRATIVE FORM

PRINCIPAL INVESTIGATOR

First Name:

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Current Role

Current Role:

PostDoc

Institution:

Italian institutions

Type of Institution:

Public research entities

Name:

130 -- Istituto Nazionale di Fisica Nucleare

Other Institution:

VAT CODE:

84001850589

Are you a permanent employee from your Home Institution to the current day?

NO

Are you retired?

NO

Residence Data

Country:

ITALIA

Province:

Roma - 058

City:

Grottaferrata

Address:

Via Domenico Petochi 26

REQUIREMENTS

Grant:
Starting Grant

Qualification

Did you obtain your first PhD degree (or other equivalent qualification) or medical specialization (for the LS macro-sector) in Italy?

NO

Foreign institution:

Technical University of Denmark, Lyngby, Copenhagen, Denmark

Have you obtained the academic recognition of your degree? Please note that the information about the academic recognition is required only if the degree was not obtained in Italy.

NO

☑ I undertake to submit the academic recognition of my degree obtained abroad promptly and within the deadline indicated by the MUR (only if the project is eligible for funding)

Did you obtained your first PhD (or equivalent qualification) or medical specialization for the LS macro-sector at least 2 (two) years and no more than 7(seven) from the expiry date of the announcement?

YES

Date of achievement of the above degree

23/11/2021

Maternity leave

Maternity leave (+18 months for each child):

n.d

Paternity leave

Paternity leave (actual amount of paternity leave taken):

n.d

Long-term illnesses, or national service

Long-term illnesses (over 90 days), or national service:

n.d

Clinical training

Clinical training (total clinical training period):

n.d

Asylum request

Asylum request (total days the PI was unable to work):

n.d

HOST INSTITUTION

Type of institution:

Public research entities

Name:

130 -- Istituto Nazionale di Fisica Nucleare

Fiscal Code / Tax ID:

84001850589

VAT Code:

04430461006

Pec:

lab.naz.frascati@pec.infn.it

Host status:

EPR

Street, Number, City, District and Country:

Via Enrico Fermi, 54,Frascati (Roma),RM,ITA

PROJECT DETAILS

Title:

Metal Oxidation and Spin state Analysis with X-ray Emission/Absorption Spectroscopy for Operando Battery Applications

Acronym:

MOSES

Grant:

Starting Grant

Project Duration:

60

Main ERC:

PE4 - Physical and Analytical Chemical Sciences

Other ERC fields 1:

Other ERC fields 2:

ERC subfields 1:

PE4_2 - Spectroscopic and spectrometric techniques

ERC subfields 2:

ERC subfields 3:

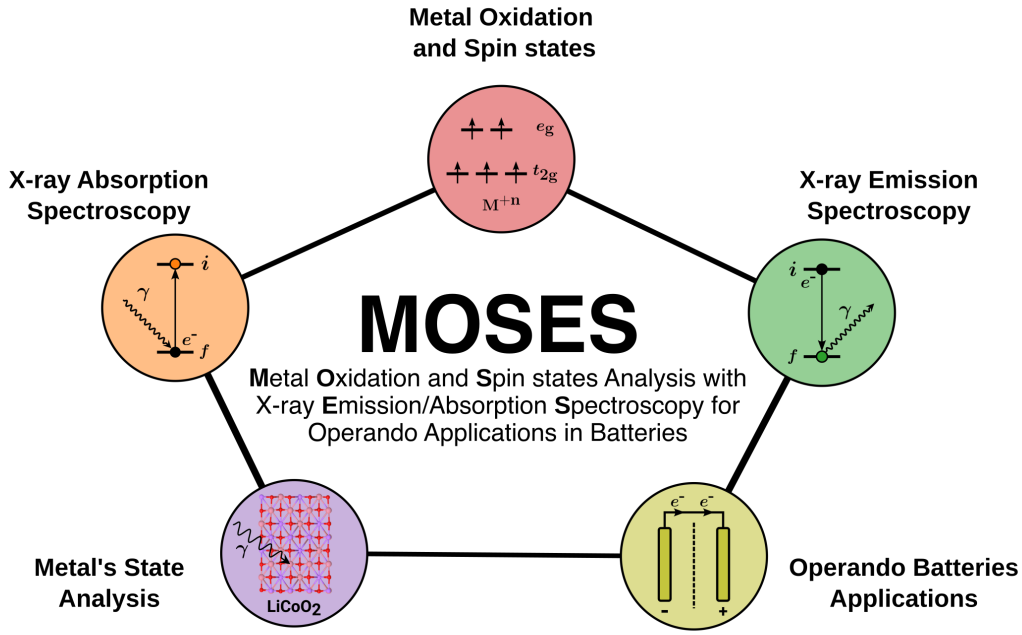
Keywords:

CHARACTERIZATION METHODS OF MATERIALS
CHEMICAL PHYSICS
ENERGY STORAGE

MACHINE LEARNING
OPTIMIZATION
SPECTROSCOPIC AND SPECTROMETRIC TECHNIQUES

Other Keywords:

Proposal Abstract:



The development of sustainable and high-performance energy storage materials is crucial for advancing the global transition to electric technologies. The properties of such materials are largely dictated by the electronic states of their constituent elements, particularly the **oxidation and spin states of metal atoms**. Consequently, there is growing attention on the ability to determine these states in real time, especially when the material is operating under working conditions.

The **MOSES project seeks to design and implement a cutting-edge prototype setup that integrates X-ray Absorption Spectroscopy (XAS) and X-ray Emission Spectroscopy (XES) to enable real-time tracking of metal oxidation and spin states**. A core objective of MOSES is to validate its capabilities through operando measurements of metal cathodes in **Li-ion batteries**, demonstrating the system's ability to analyze materials in realistic operational environments.

To achieve real-time sensitivity to changes in the metal state under operando conditions, the MOSES setup will be optimized for high sensitivity and minimized measurement time. The fully automated system will rapidly align its components, maximizing the Signal-to-Background Ratio (SBR) of the acquired spectra. **MOSES will accomplish this by developing an advanced optimization framework based on a Digital Twin of the experiment**. This framework will enable automatic optimization of setup parameters to enhance the detection of the absorption edge and emission line under study. **The approach will use a model-based Reinforcement Learning algorithm trained on simulated experiments, performed by combining Ab Initio calculations and Ray-Tracing simulations** to fully replicate the observed XAS and XES spectra.

By significantly improving the ability to track dynamic changes in the electronic configuration of metals, the **MOSES** project is poised to improve battery characterization techniques, potentially accelerating the design of next-generation energy storage materials. Beyond batteries, the methodology established by **MOSES is expected to have wide-ranging applications in materials science, catalysis, and advanced manufacturing**, facilitating in-depth studies of metal oxidation and spin states across diverse technological domains.

BUDGET

Expenditure item	Euros
A. Principal Investigator:	309.600,00
B.Personnel :	371.520,00
C.Instruments and equipment:	96.000,00
D.Other operational costs:	312.200,00
E.Scientific and technical consulting services:	0,00
F. Overheads:	136.224,00
Total Cost:	1.225.544,00

OTHER FUNDING AND RELATED PI TIME COMMITMENT EXTERNAL TO THE PROJECT

Proposal Title	Funding Source	Amount (Euro)	Start Date	End Date	PI Role	Relation to current Proposal	Time, in percentage terms dedicated to the project

TEACHING AND MANAGEMENT COMMITMENTS EXTERNAL TO THE PROJECT

Institution	Start Date	End Date	Role	Percentage time dedicated to teaching and management commitments external to the project

SECTION B - SCIENTIFIC PROPOSAL

SYNOPSIS

Main ERC Field:

PE4 - Physical and Analytical Chemical Sciences

Other ERC Field 1:

Other ERC Field 2:

ERC Subfield 1:

PE4_2 - Spectroscopic and spectrometric techniques

ERC Subfield 2:

ERC Subfield 3:

PROPOSAL SUMMARY

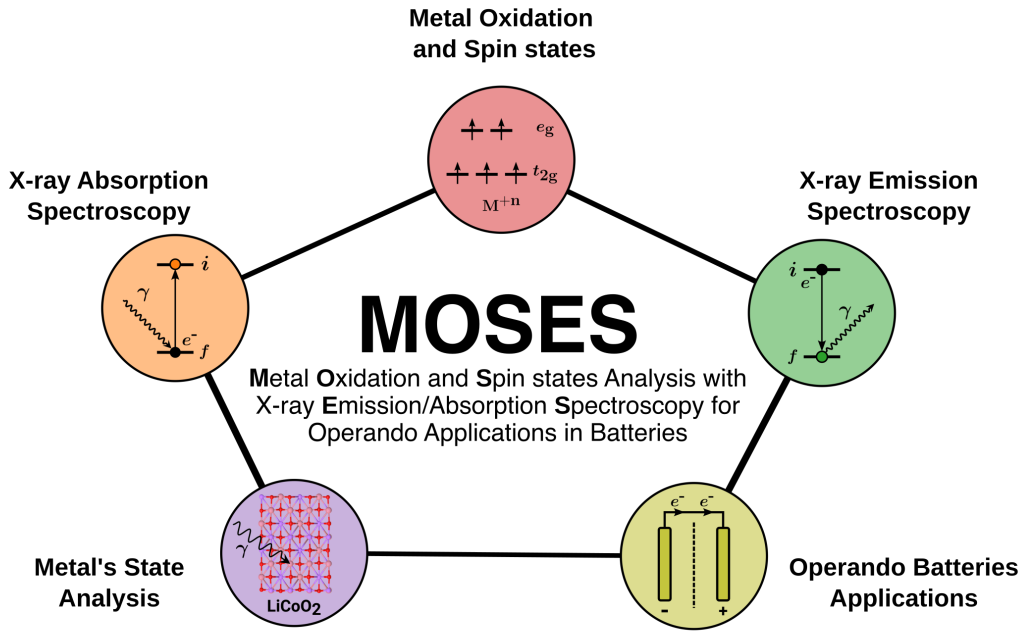


FIGURE B1: MOSES Key Aspects. The MOSES project aims to investigate metal oxidation and spin states through X-ray Absorption Spectroscopy (XAS) and X-ray Emission Spectroscopy (XES), offering complementary insights into the electronic structure. A prototype setup will be designed, integrating both techniques into a fully automated and motorized system. Operando measurements on a working battery will serve as primary test-case application to showcase setup's capabilities.

The **global transition to sustainable energy and electric technologies is accelerating**, particularly in the energy and automotive sectors, with investments reaching a record $2.1 trillion in 2024, an 11% increase from the previous year [1].

Achieving this global transition at a large scale, necessitates advanced energy materials that offer enhanced efficiency, durability, and environmental compatibility. For energy storage systems, such as **Lithium-Ion Batteries (LIBs)**, the urgent challenge lies in identifying environmentally sustainable materials that enhance efficiency and lifespan. Research efforts predominantly focus on identifying cathode metals that enhance performance, ensure resource availability, and reduce ecological impact. The electronic states of these metals play a crucial role in Li-ion migration, directly affecting battery performance and longevity. Beyond LIBs, oxidation and spin states critically influence material properties in various applications, including catalysis for green hydrogen production and carbon capture technologies.

Precise monitoring of these states under real-time operating conditions is essential, extending its significance beyond battery applications to various technological fields. X-ray techniques are widely employed for such investigations, with the goal of performing an **operando spectroscopy**, a real-time analysis of metal states within the operational environment of the material under study.

Within this context **(see FIGURE B1)**, the **MOSES** project aligns with the ERC field **PE4 (Physical and Analytical Chemical Sciences)** and specifically subfield **PE4_2 (Spectroscopic and Spectrometric Techniques)**, and aims to develop:

> **MAIN OBJECTIVE: MOSES** will develop **a cutting-edge, fully automated prototype setup** for analyzing **spin and oxidation states of metals using X-ray Absorption/Emission spectroscopy.** This approach will enable **real-time tracking of metal's electronic state,** enhancing applications **in operando battery research.**

Among the various available techniques, **X-ray spectroscopy is particularly advantageous for operando characterization of metal states due to its direct sensitivity to both oxidation and spin states.** Furthermore, compared to infrared or optical methods, **X-ray techniques offer enhanced sensitivity and superior penetration depth**, enabling more precise and reliable monitoring under real operational conditions [2].

Specifically, **MOSES** will utilize **X-ray Absorption Spectroscopy (XAS) [3] and X-ray Emission Spectroscopy (XES) [4] (see FIGURE B2)** to provide complementary, detailed insights into the electronic properties under operando conditions for transition metal complexes. **XAS** will specifically probe the **oxidation state** of the metal, while **XES** will enable accurate determination of electronic transitions and **spin states**, enhancing our understanding of the material's dynamic behavior in real-time operation.

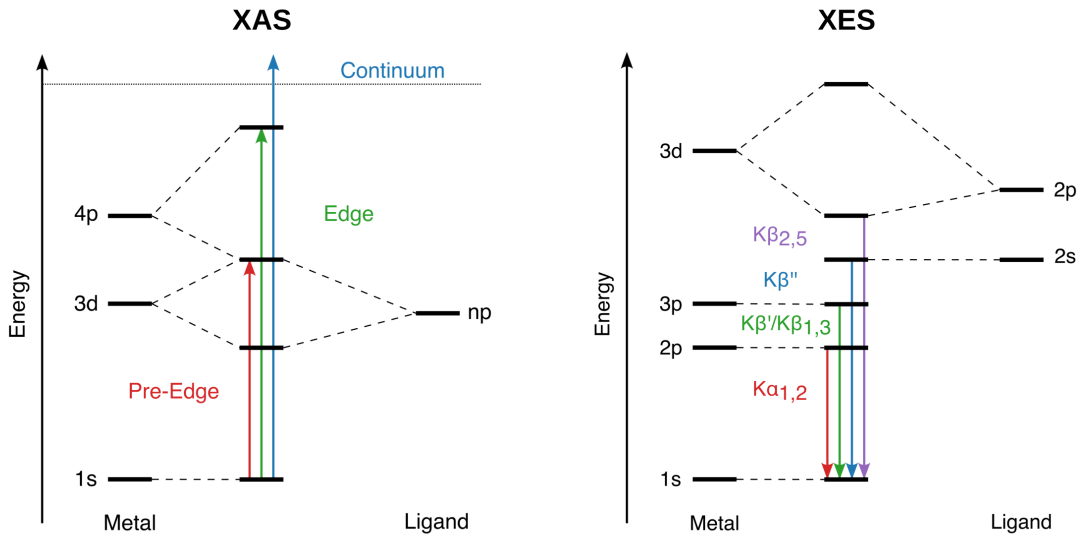


FIGURE B2: Metal-Ligand transitions for XAS and XES. A comparison of the transitions involved in X-ray Absorption Spectroscopy (XAS) and X-ray Emission Spectroscopy (XES) for a transition metal complex. XAS probes excitations from the 1s core level to unoccupied electronic states, revealing pre-edge, edge (XANES), and continuum features (NEXAFS), which provide insights into oxidation state and coordination environment. Conversely, XES captures radiative transitions from occupied states back to the 1s core level, yielding characteristic Kα fine structure and Kβ satellite peaks that encode information on spin state and electronic interactions.

OBJECTIVES

Operando measurements with extremely high temporal resolution require an XAS and XES setup optimized for increased sensitivity, specifically enhancing the **Signal-to-Background Ratio (SBR)** of the absorption edge or emission line under investigation [5]. This high sensitivity is essential for accurately capturing subtle variations in the metal's electronic state, necessitating an exceptionally low **Limit of Detection (LOD)**.

Performing durability tests, for example, requires several alignment and optimization procedures, necessitating a fully automated and motorized setup capable of autonomous optimization. Modern methods employing **Machine Learning (ML)**, such as **Reinforcement Learning (RL)**, offer these possibilities, yet there is a lack of applications of such approaches to X-ray experiments [6].

Additionally, while optimization can be performed using black-box methods, simulations specifically targeting the measured quantities can enhance both the quality and precision of the optimization process. Simulating XAS and XES spectra through **ab initio methods** is a well-established approach [7]. However, it remains uncommon to use such simulations to directly enhance experimental outputs. Achieving this integration requires developing a comprehensive pipeline that incorporates ab initio spectral calculations alongside sophisticated ray-tracing simulations, ultimately refining the measurement strategy and data quality.

The **Key Objectives of MOSES** aim to address these research questions and are outlined in the following in relation to their **Work Package (WP)**, summarized in **FIGURE B3**.

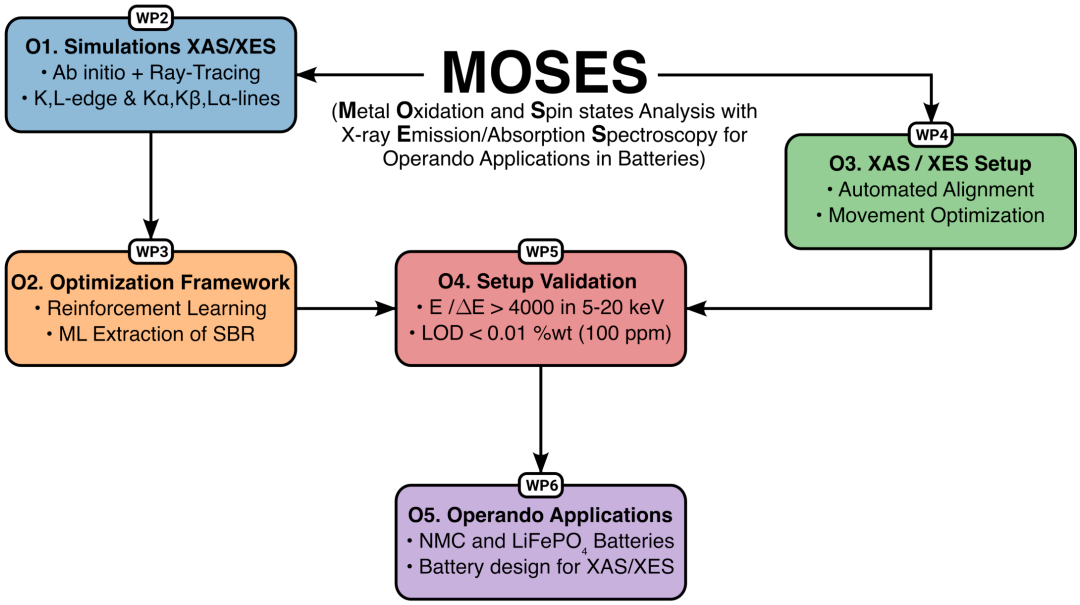


FIGURE B3: Key Objectives of MOSES. Schematic representation of the MOSES's key objectives, starting from the simulations for XAS and XES spectra (O1), the machine learning-driven optimization framework (O2), building the automated experimental setup (O3), the rigorous validation of the developed system (O4), and operando spectroscopy applications for real-time monitoring of metal states in battery research (O5). Each objective is related to the respective Work Package (WP).

OBJECTIVE 1 (O1): I will integrate advanced ab initio computational methods with ray-tracing simulations (WP2) to comprehensively model the K and L absorption edges for XAS, and the Kα, Kβ, and Lα emission lines for XES. This integration aims to achieve a **prediction accuracy within 1%** when compared with experimental spectra.

OBJECTIVE 2 (O2): I will establish and deploy **an advanced optimization framework employing ML techniques (WP3) trained on simulations obtained from WP2 methodologies**. Specifically, I will use RL to iteratively refine setup parameters, **aiming for a minimum tenfold improvement in the SBR for the target emission or absorption edge**. The effectiveness of the optimization approach will be rigorously validated by comparison with conventional grid scan techniques.

OBJECTIVE 3 (O3): I will design and build **a fully automated prototype setup for simultaneous XAS and XES**, with motorized control of all critical components, including the X-ray source, monochromator crystals, and detectors **(WP4)**. The automation will enable precise alignment with the specific energy regions of interest, **facilitate simultaneous XAS and XES measurements**, and optimize the fine movements of all components **to maximize the SBR**.

OBJECTIVE 4 (O4): I will apply the **framework developed in WP4 to the automated and combined XAS/XES setup (WP5)**. This will include fine-tuning the setup components' positions and angles to maximize the SBR, achieving a **LOD below 0.01 wt% (< 100 ppm)**. The setup will maintain a **high energy resolution (E / ΔE > 4000) for metals with edges/lines in the 5-20 keV range** and complete measurements **within an acquisition time on the order of minutes**.

OBJECTIVE 5 (O5): I will show the potentiality of the optimized XAS/XES setup through **an operando study of Li-ion battery cathodes, focusing on NMC ($LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$) and $LiFePO_4$**, whose K-edges and Kα and Kβ emission lines fall within the 5-20 keV range. **A battery prototype with optimized cathode thickness will be designed to enhance X-ray sensitivity for accurate oxidation and spin state tracking**. The study will enable **real-time monitoring during charge/discharge cycles with a 10-minute temporal resolution**.

METHODOLOGY

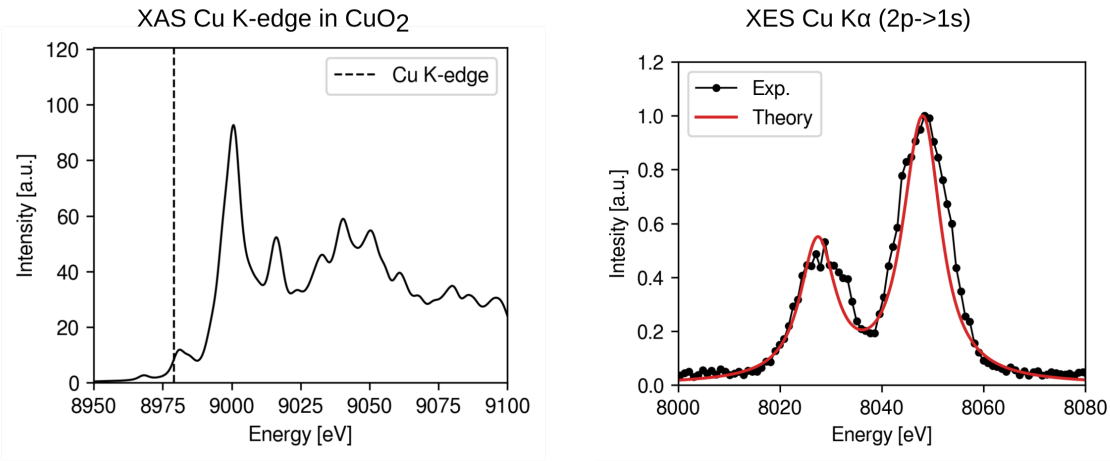


FIGURE B4: Ab initio Calculations for XAS and XES. Example calculations from first principles for the absorption K-edge for copper in CuO_2 (left) taken from Material Project (https://next-gen.materialsproject.org/materials/mp-14549?formula=CuO) with reference K-edge value with a dashed lines and on the right the Kα (2p->1s) emission line for an isolated copper atom (red) in comparison with the experimental XES spectrum (black dots) obtained with the VOXES setup at LNF.

MOSES is structured into **six Work Packages (WPs)**: five dedicated to the project's key objectives **(WP2-WP6)**, and one focused on **personnel recruitment (WP1)**, where three researchers will be hired to support the project.

- **WP2:** Ab initio simulations will performed with **Density Functional Theory (DFT) using the software GPAW [8]**, which I have extensively applied in my research to compute absorption and emission spectra. **GPAW includes a module for absorption edge calculations**; however, for XES, where correlation effects play a more significant role, I plan to use the **Multiconfigurational Dirac-Fock (MCDF) method to accurately capture multiplet effects**. **These lineshape calculations (see FIGURE B4) will serve as inputs for the Ray-Tracing modeling**, enabling a comprehensive simulation of absorption edge and emission lines as detected by the experimental setup, utilizing the **XRT package [9]**.

- **WP3: The workflow of the optimization framework of WP3 is outlined in FIGURE B5. This framework will employ a model-based RL approach, exploiting simulated spectra from WP2 for pretraining. By establishing an RL feedback loop, the algorithm will iteratively refine the experimental setup parameters, optimizing the SBR based on the learned policy.** A critical component of this optimization framework is the **ML model designed to accurately extract the SBR from the spectra**. This task will be accomplished using a **Conditional Variational Autoencoder (CVAE)**, trained on simulation data of WP2. **The CVAE was also chosen to have the possibility of learning over historical data**, and further enhance its SBR extraction capabilities by generating augmented spectra, thereby bridging the gap between simulated and experimental spectra. **Here, my previous experience combining ab initio method and ML will valuable and totally transferrable** [10].

- **WP4:** The **fully automated and motorized prototype setup of WP4** is illustrated in **FIGURE B6**. It comprises two combined lines: one for **XAS in transmission mode** and one for **XES** in **fluorescence mode**. Each line integrates slits, mosaic crystals, and a pixel detector, automatically aligning and optimizing its SBR independently.

- **WP5:** The system's automated parameter optimization will be thoroughly validated by comparing its results **against a traditional grid-scan approach of the relevant parameters**.

- **WP6:** For the operando battery application showcase, a specialized battery cell will be designed, featuring a window made from Kapton or, preferably, beryllium, allowing direct exposure of both cathode and anode. **This design will be supported by simulations of X-ray penetration to ensure optimal sensitivity for conducting XAS and XES analyses**. These simulations will determine the ideal thickness and configuration to maximize measurement accuracy. **Lithium-ion battery cathode materials**, specifically **NMC and Fe-based cathodes**, will be investigated due to their suitable metal edges (Ni, Mn, Co, and Fe) falling within the operational energy range of the setup (5-20 keV).

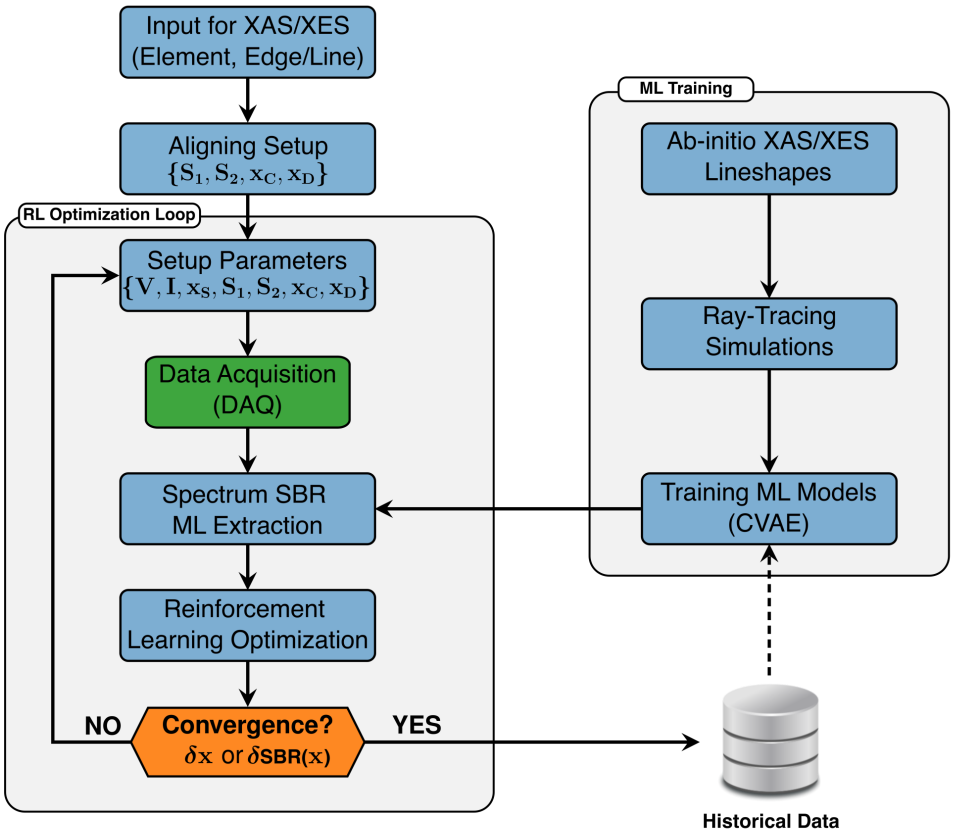


FIGURE B5: MOSES Optimization Workflow. The optimization workflow involves several key steps. First, the framework aligns the setup based on the selected input absorption edge or emission line and initiates the measurement with the DAQ, where $\mathbf{x_C}$ and $\mathbf{x_D}$ are collective coordinates for crystals and detectors [see FIGURE B6 of the setup]. The Signal-to-Background Ratio extraction is done using Machine Learning (CVAE), previously trained on the ray-tracing simulations. The optimization loop continues until convergence, depending on the threshold over the parameters x or on the SBR, then the results are saved in a database, that eventually can be used for training the CVAE on historical real data.

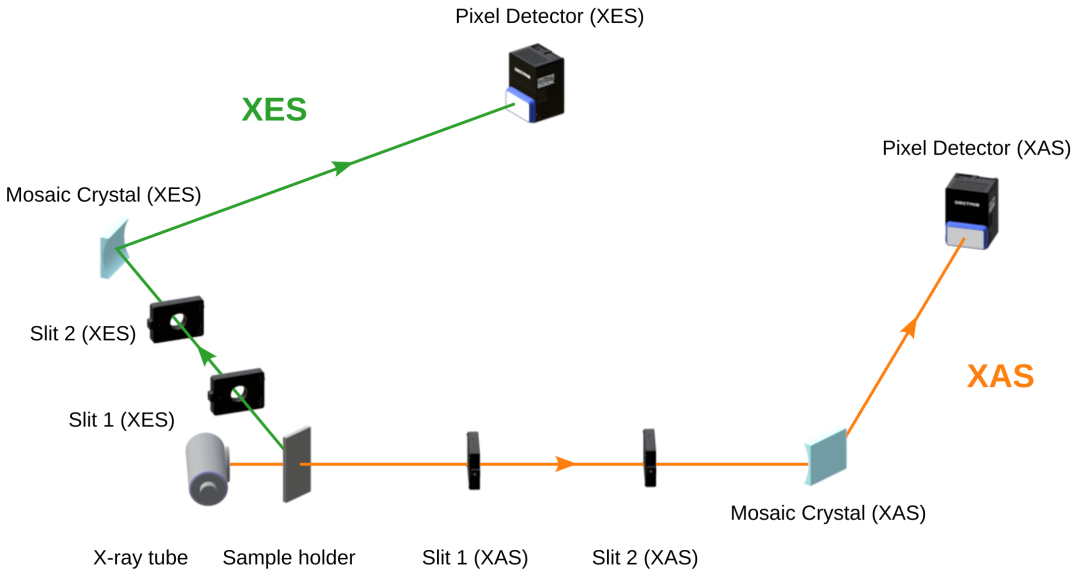


FIGURE B6: Prototype Setup for combined XAS and XES. The proposed setup for the MOSES project comprises an X-ray tube, a sample holder, and two lines: one for XAS (orange) and one for XES (green). Each line includes two slits, a mosaic crystal, and a pixel detector. Mosaic crystals and pixel detectors will have motorized stages for aligning to the edge/line of interest. Addiotionaly, together with the sample holder, they will also have axis for fine aligment to optimize the signal-to-background ratio of the spectra.

TEAM AND INFRASTRUCTURE

MOSES will be conducted at the **National Laboratories of Frascati (LNF) of INFN**, involving the Principal Investigator (PI) and three researchers **(see Gantt chart in FIGURE B7)**.

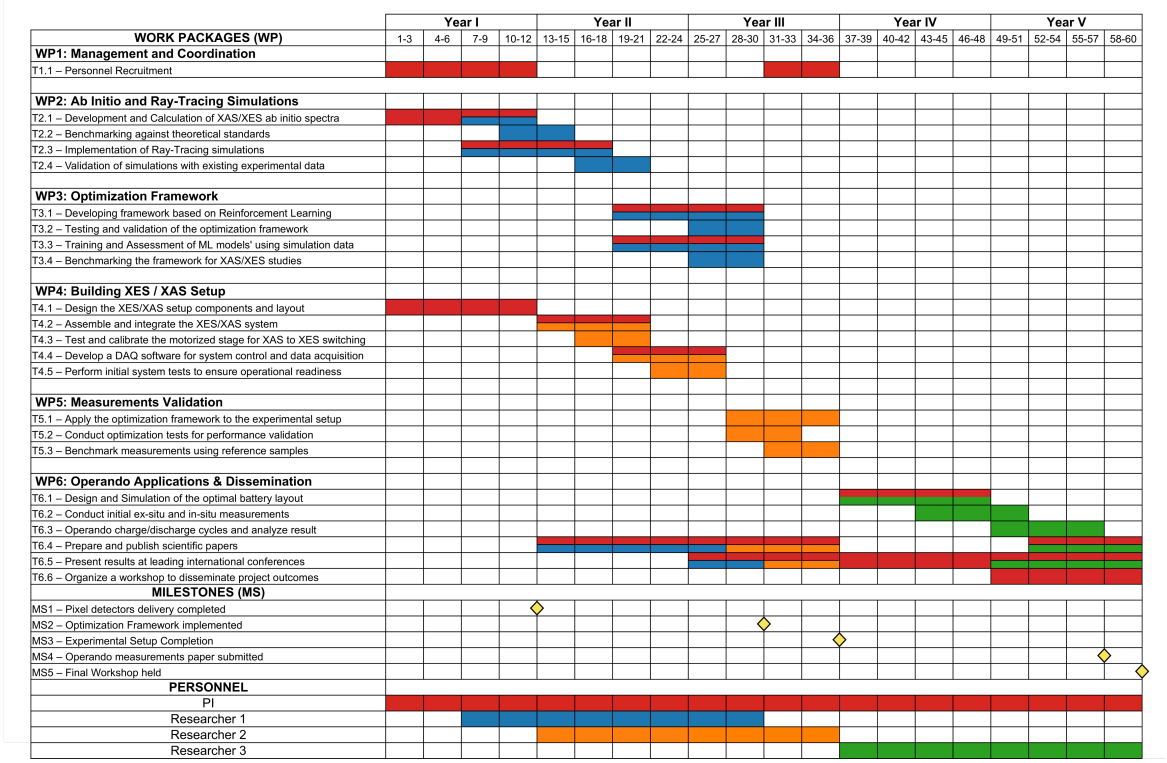


FIGURE B7: Gantt chart of the MOSES project. This Gantt chart illustrates the timeline of the MOSES project, spanning five years and organized by Work Package (WP) and Task (T). Project Milestones (MS) are indicated along with their expected release dates. The Gantt chart also includes personnel involvement, represented with individual colors for each team member. Tasks involving multiple contributors are highlighted using a combination of colors to reflect collaboration.

EXPECTED IMPACT

MOSES will advance operando X-ray spectroscopy, benefiting multiple sectors **(see FIGURE B8)**:

- **Scientific:** Pioneering real-time characterization techniques for energy materials.
- **Technological:** Developing a high-precision, fully automated spectroscopy system.
- **Industrial:** Enhancing battery and catalyst research, directly supporting the energy transition.
- **Societal:** Contributing to sustainable energy storage and cleaner energy solutions.
- **Environmental:** Reducing material waste and optimizing battery lifespans.

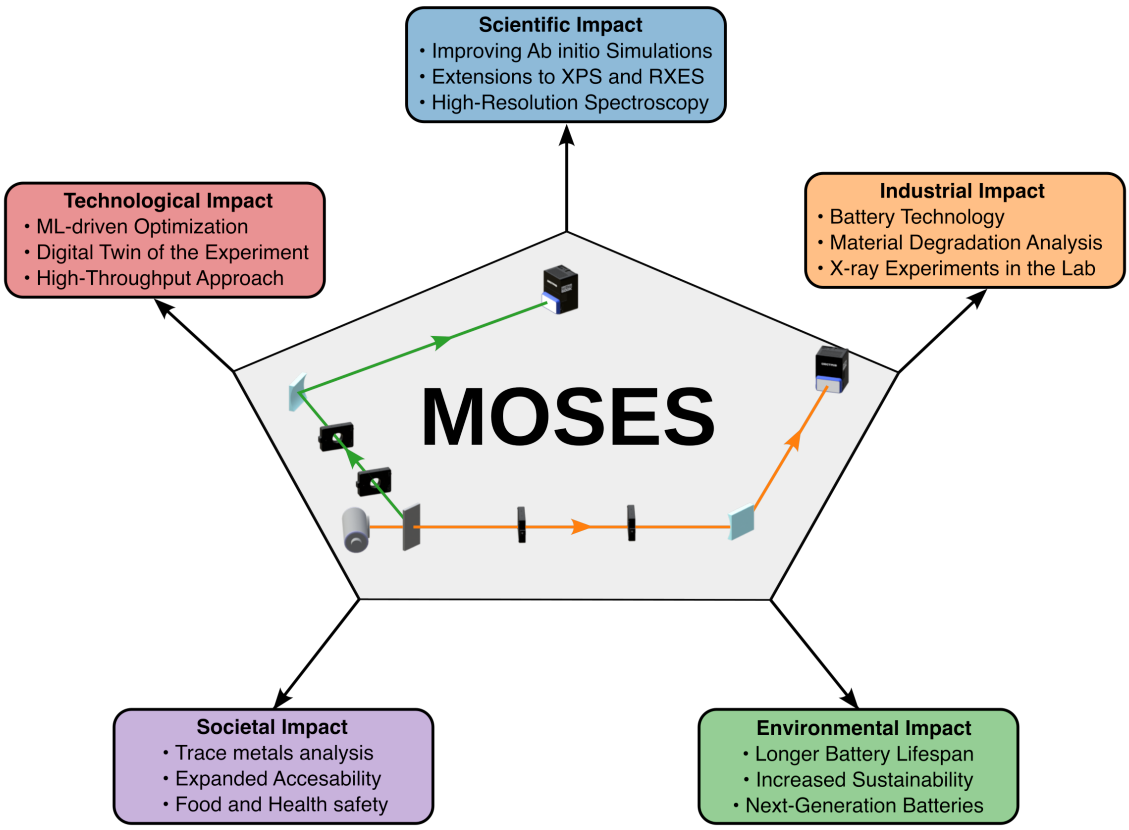


FIGURE B8: MOSES Project Impact Overview. The MOSES project advances X-ray spectroscopy with impacts in science, technology, industry, society and environment.

CONCLUSION

The successful realization of the main objective of MOSES will represent a breakthrough in X-ray operando spectroscopy of metals, aligning with the global push for sustainable and electric technologies. By integrating **ab initio simulations, experimental techniques, and advanced ML methods**, **MOSES will accelerate innovation in energy storage and catalysis**, addressing pressing technological challenges at a crucial moment for global sustainability efforts.

REFERENCES

[1] BloombergNEF. 2025. "Global Investment in the Energy Transition Exceeded $2 Trillion for the First Time in 2024, According to BloombergNEF Report." January 30, 2025.

[2] https://doi.org/10.1039/D1CP03263A

[3] https://doi.org/10.1017/CBO9780511809194

[4] https://doi.org/10.1088/0268-1242/29/2/023002

[5] https://doi.org/10.1016/j.ccr.2020.213466

[6] https://doi.org/10.1088/2632-2153/ad4253

[7] https://doi.org/10.1201/9781420008425

[8] https://doi.org/10.1063/5.0182685

[9] https://doi.org/10.1117/12.2061400

[10] https://doi.org/10.1038/s41524-023-00977-x

CURRICULUM VITAE

Personal Information:

Family Name: Simone
First Name: Manti
Researcher unique identifier(s) (such as ORCID, Research ID, etc. ...): https://orcid.org/0000-0003-3770-0863
Date of birth: 18/11/1993
Nationality: Italian

Education:

2021 - PhD in Physics
Faculty Physics, Technical University of Denmark, Denmark
2017 - Master
Faculty Physics, University of Rome “Tor Vergata”, Italy
2015 - Bachelor
Faculty Material Science, University of Rome “Tor Vergata”, Italy

Current Position(s):

2022 - Assegnista
INFN Laboratories of Frascati (LNF), Italy

Previous Positions:

2021-2022 Research Assistant
Faculty of Physics, Technical University of Denmark, Denmark

Fellowships and awards:

2017 - First in the ranking of merit of second-year students of the Masters degree of Physics, Univeristy of Roma Torvergata, Rome, Italy

SUPERVISION OF GRADUATE STUDENTS; PhD POSTDOC STUDENTS:

2024 - Supervision of Summer School student (Bando 26508/24)
LNF INFN, Frascati, Italy

TEACHING ACTIVITIES:

2019 - Teaching Assistant - Electronic Structure Methods in Material Physics (10302), Technical University of Denmark
2018 - Teaching Assistant - Condensed Matter Physics and Nanoscale Materials Physics (10303), Technical University of Denmark

ORGANISATION OF SCIENTIFIC MEETINGS:

2023 - Organization and chair of conference ALPACA: modern ALgorithms in machine learning and data analysis from medical physics to research with ACcelerAtors and in underground laboratories, ECT*, Trento, Italy
2025 - Organization and chair of confenrence HPXM2025, LNF, Frascati, Italy

INSTITUTIONAL RESPONSIBILITIES:

NONE

REVIEWING ACTIVITIES:

Peer reviewer for the journal Few-Body Systems

MEMBERSHIPS OF SCIENTIFIC SOCIETIES:

NONE

MAJOR COLLABORATIONS:

Dr. C. Curceanu, Laboratori Nazionali di Frascati INFN, spokesperson of the SIDDHARTA-2 experiment: Kaonic atoms measurements
Prof. P Indelicato, Research Director at Laboratoire Kastler Brossel (CNRS), France: exotic atoms calculations

Prof. Lajos Diosi, Wigner Research Centre for Physics, Hungary, Collapse models

TRACK RECORDS

Track Record:

Publications as lead author:

1. Manti S, Svendsen MK, Knøsgaard NR, Lyngby PM, Thygesen KS. Exploring and machine learning structural instabilities in 2D materials. Npj Comput Mater. 2023 Mar 4;9(1):1–10.
2. Manti S, Napolitano F, Clozza A, Curceanu C, Moskal G, Piscicchia K, et al. Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations. Condens Matter. 2024 Mar;9(1):19.
3. Manti S, Bazzi M, Bortolotti N, Capoccia C, Cargnelli M, Clozza A, et al. Testing the Pauli Exclusion Principle across the Periodic Table with the VIP-3 Experiment. Entropy. 2024 Sep;26(9):752.

Selection of publications as co-author:

1. Riis-Jensen AC, Simone Manti, Thygesen KS. Engineering Atomically Sharp Potential Steps and Band Alignment at Solid Interfaces using 2D Janus Layers. J Phys Chem C. 2020;124(17):9572–80.
2. Moustafa H, Kangsabanik J, Bertoldo F, Manti S, Thygesen KS, Jacobsen KW, Olsen T. Selenium and the role of defects for photovoltaic applications. Phys Rev Mater. 2024 Jan;8(1):015402.
3. Gjerding MN, Taghizadeh A, Rasmussen A, Ali S, Bertoldo F, Deilmann T, Knøsgaard NR, Kruse M, Larsen AH, Manti S, Pedersen TG, Petralanda U, Skovhus T, Svendsen MK, Mortensen JJ, Olsen T, Thygesen KS. Recent progress of the computational 2D materials database (C2DB). 2D Mater. 2021 Jul;8(4):044002.
4. Ali S, Nilsson FA, Manti S, Bertoldo F, Mortensen JJ, Thygesen KS. High-Throughput Search for Triplet Point Defects with Narrow Emission Lines in 2D Materials. ACS Nano. 2023;17(21):21105–15.
5. Bertoldo F, Ali S, Manti S, Thygesen KS. Quantum point defects in 2D materials - the QPOD database. Npj Comput Mater. 2022 Apr 4;8(1):56.
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Invited talks:

1. 11th International Conference on Nuclear Physics at Storage Rings (STORI24), Huizhou, China, November 2024
Invited talk: “Unlocking the Secrets of Kaonic Atoms with the DAFNE Collider: from DEAR to SIDDHARTA-2”
2. KAMPAI - Kaonic, Antiprotonic, Muonic, Pionic and onia exotic Atoms: Interchanging knowledge and recent results, Trento, Italy, October 2024
Invited talk: “Kaonic Atom Properties and Cascade Models: Exploring Strong Interactions through First-Principles Calculations and Monte Carlo simulations”
3. Advances in the Investigation of Weak and Strong Interactions, Bucharest, Romania, July 2024
Invited talk: “Kaonic Atom Properties and Cascade Models: Exploring Strong Interactions through First-Principles Calculations”
4. A Modern Odyssey: Quantum Gravity meets Quantum Collapse at Atomic and Nuclear Physics Energy Scales in the Cosmic Silence, ECT*, Trento, Italy, June 2024
Invited talk: “Cancellation Effects in the Spontaneous Collapse Rate at Low Energy Limit”
5. ALPACA: Modern Algorithms in Machine Learning and Data Analysis from Medical Physics to Research with Accelerators and in Underground Laboratories, ECT*, Trento, Italy, November 2023
Invited talk: “ML in SIDDHARTA-2: The Monitoring Challenge”
6. EXOTICO: Exotic Atoms Meet Nuclear Collisions for a New Frontier Precision Era in Low-Energy Strangeness Nuclear Physics, ECT*, Trento, Italy, October 2022
Invited talk: “Cascade Models for Atomic Transitions in Kaonic Atoms”

Visiting Scientist:

- Wigner Research Centre for Physics, Budapest, Hungary Supervisor: Lajos Diosi, January 2024
- Sorbonne Université, Paris, France Supervisor: Paul Indelicato, December 2023
- Center for Physical Sciences and Technology (FTMC), Vilnius, Lithuania Supervisor: Audrius Alkauskas, September 2019 – December 2019

Description of main research results:

Throughout my research career, I have extensively applied ab initio simulations and experimental methods to investigate materials and fundamental interactions, achieving significant outcomes in several areas:

• During my Ph.D. at the Technical University of Denmark (DTU), I focused on the electronic and optical properties of two-dimensional (2D) materials. My work included extensive use of Density Functional Theory (DFT) simulations for evaluating dynamical stability, electronic structure, and optical responses. I combined these simulations with machine learning techniques to enhance predictive capabilities, contributing to databases such as the Computational 2D Materials Database (C2DB) and the Quantum Properties of Defects (QPOD). Furthermore, I explored defect-induced optical properties in 2D materials, with implications for single-photon emitters and quantum computing applications.

• At the National Laboratory of Frascati (LNF) of INFN, within the SIDDHARTA-2 collaboration, I performed Multi-Configuration Dirac-Fock (MCDF) simulations to calculate electronic transitions in exotic atoms, specifically kaonic atoms. These calculations provided essential theoretical support for interpreting experimental data on kaonic atoms, improving our understanding of strong interaction effects in exotic atomic systems.

• Within the MITIQO project at LNF, I optimized X-ray fluorescence (XRF) experimental setups under the VOXES initiative, specifically targeting the detection of metal content in edible liquids such as wine and olive oil. By integrating machine learning algorithms, I significantly enhanced analytical capabilities, enabling automated classification of samples by metal content and improving the sensitivity and reproducibility of the measurements, demonstrating a valuable application of advanced analytical techniques in food quality control.

• Additionally, my research at the National Laboratory of Gran Sasso contributed to the VIP experiment, where I utilized MCDF calculations to predict violation of Pauli Principle in X-ray emission spectra. Additionally, I participated in experiments on spontaneous wavefunction collapse to enhance the description of collapse models through ab initio simulations

DESCRIPTION OF ACTIVITIES CONDUCTED DURING THE PhD

Project Title:	institution:	Start Date:	End Date:	Activity description:	PI Role:
C2DB	Technical University of Denmark	14/05/2018	14/09/2021	Machine Learning on 2D materials ab initio simulations	Project Contributor
QPOD	Technical University of Denmark	14/05/2018	14/09/2021	Optical properties of defect in 2D materials	Collaborator

SECTION C

RESUME GENERAL INFORMATION

Project Title:

Metal Oxidation and Spin state Analysis with X-ray Emission/Absorption Spectroscopy for Operando Battery Applications

Acronym:

MOSES

Principal Investigator Name:

SIMONE MANTI

Name of the project's host institution:

130 -- Istituto Nazionale di Fisica Nucleare

Duration of the project - X month:

60

SCIENTIFIC PROPOSAL

State of the art and objectives:

The advancement of materials for energy applications is directly linked to the electronic properties of the metal elements involved. As noted John B. Goodenough, co-winner of the 2019 Nobel Prize for the development of **Lithium-Ion Batteries (LIBs)**, highlighted over five decades ago [1], **the performance of these materials is fundamentally determined by the metal's magnetic and chemical environment**. In LIBs, for instance, cobalt's oxidation and spin states crucially dictate its effectiveness as a cathode material, thereby directly influencing lithium-ion mobility throughout charge and discharge cycles. Similarly, in **catalytic applications**, the metal's electronic structure governs key processes such as hydrogen migration. These cases underscore **the essential connection between a metal's electronic properties and its functional role**, not only in energy-related materials but also across a broader spectrum of chemical and materials science disciplines.

Within this context **(see FIGURE C1)**, the **MOSES project** aims to advance this field further with the main objective of:

> **MAIN OBJECTIVE: MOSES** will develop a **cutting-edge, fully automated prototype setup** for analyzing **spin and oxidation states** of metals using **X-ray Absorption/Emission spectroscopy**. This approach will allow **real-time tracking of metal cathode's electronic state**, enabling applications in **operando battery research**.

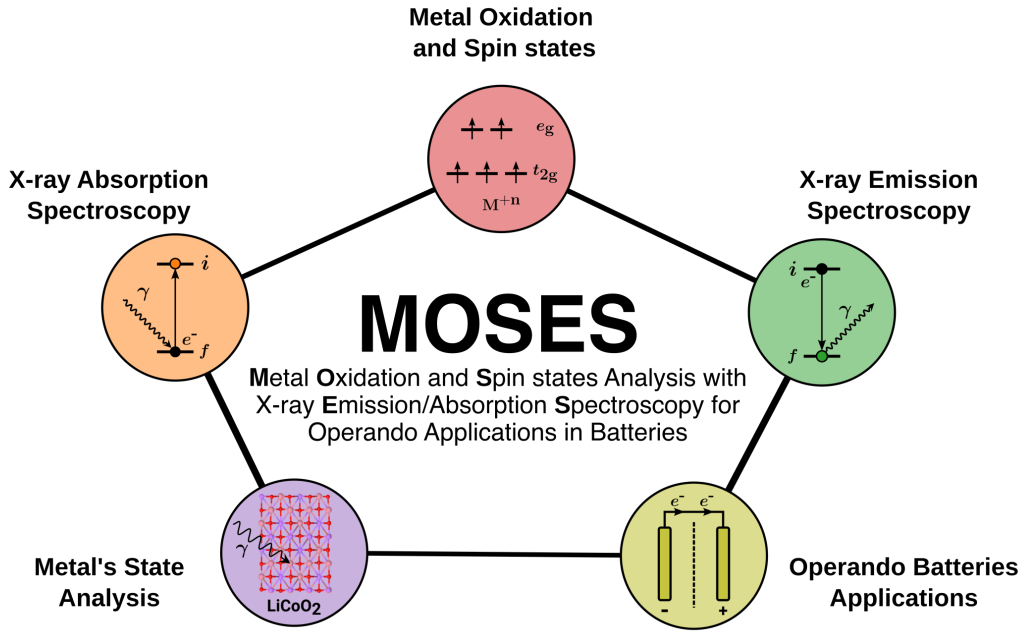


FIGURE C1: MOSES Key Aspects. The MOSES project aims to investigate metal oxidation and spin states through X-ray Absorption Spectroscopy (XAS) and X-ray Emission Spectroscopy (XES), offering complementary insights into the electronic structure. A prototype setup will be designed, integrating both techniques into a fully automated and motorized system. Operando measurements on a working battery will serve as primary test-case application to showcase setup's capabilities.

In the following, I will outline the **Key Objectives (O1-O5)** of **MOSES** and **highlight how they address critical gaps and limitations in the current state of the art.**

X-ray spectroscopy is an established and versatile analytical technique, widely applied in diverse fields including Materials Science, Chemistry, Physics, Architecture, Art, Archaeology, Astronomy, and Industry [2]. Compared to infrared or optical methods, X-rays offer superior penetration capability and significantly greater sensitivity to chemical environments, as well as oxidation and spin states of metals. This enhanced sensitivity makes X-ray spectroscopy uniquely powerful for investigating the detailed electronic properties of materials.

Among the various techniques utilizing X-rays, **core-hole spectroscopies** [3] are especially informative regarding the electronic states of metals. These methods involve creating a vacancy (core-hole) in the inner shell of an atom **(see FIGURE C2)**, which subsequently provides detailed insights into its electronic configuration. The analysis can focus either **on the absorption characteristics during the initial hole creation or on the emitted X-rays following the electron relaxation processes**, thus enabling precise investigation of metal oxidation states, spin states, and chemical environments.

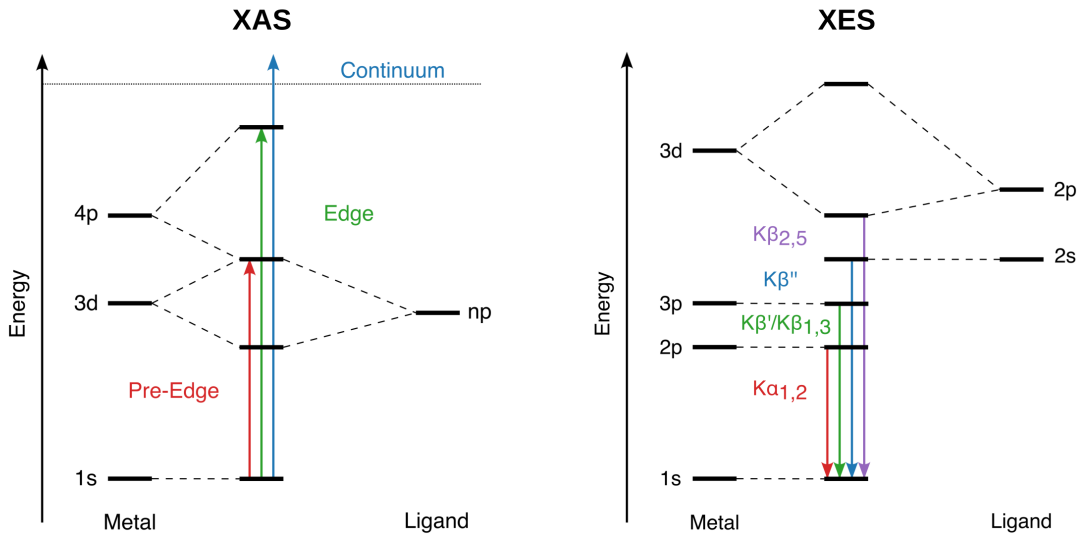


FIGURE C2: Metal-Ligand transitions for XAS and XES. A comparison of the transitions involved in X-ray Absorption Spectroscopy (XAS) and X-ray Emission Spectroscopy (XES) for a transition metal complex. XAS probes excitations from the 1s core level to unoccupied electronic states, revealing pre-edge, edge (XANES), and continuum features (NEXAFS), which provide insights into oxidation state and coordination environment. Conversely, XES captures radiative transitions from occupied states back to the 1s core level, yielding characteristic Kα fine structure and Kβ satellite peaks that encode information on spin state and electronic interactions.

X-ray Absorption Spectroscopy (XAS) [4] analyzes how a material's absorption coefficient in the X-ray range reveals information about its **unoccupied electronic states**. Specifically, for metal atoms, XAS provides detailed insights into their oxidation states and local chemical environments. Depending on which electron shell is involved, distinct absorption edges (e.g., K-edge, L-edge) can be observed. For transition metal complexes, the K-edge spectrum typically comprises two primary regions **(see FIGURE C2)**:

• **X-ray Absorption Near Edge Structure (XANES)**, spanning from approximately 50 eV below to about 100 eV above the main absorption edge, which is sensitive to oxidation states and local symmetry.

• **Extended X-ray Absorption Fine Structure (EXAFS)**, beginning just above the XANES region and extending several hundred eV beyond the main edge, providing precise information about interatomic distances and coordination numbers.

X-ray Emission Spectroscopy (XES) [5] analyzes X-rays emitted during electron recombination in core-hole states, focusing on **occupied electronic levels**. The emission lines observed depend on the electronic transitions between specific levels, and are designated according to the initial and final electron states involved. For transition metal complexes, notable emission lines include the Kα line, resulting from the recombination of electrons transitioning from the 2p to the 1s level, and the Kβ line arising from 3p to 1s transitions. **XES offers complementary insights to XAS, particularly concerning the spin state and local structural environment of metals**. Specifically, the Kβ emission lines are highly informative about the metal's spin state due to the strong interaction between 3p and 3d electrons, providing valuable insights into their electronic configuration and spin arrangement.

Both X-ray Absorption Spectroscopy (XAS) and X-ray Emission Spectroscopy (XES) are well-established from a theoretical standpoint [3]. These techniques can be described either through semi-empirical **multiplet methods** [6], or fully ab initio approaches such as **Density Functional Theory (DFT)** [7] and more sophisticated **Multiconfiguration Dirac-Fock (MCDF)** methods [8], which allow accurate predictions of spectral features and provide deep insights into electronic structure and interactions.

To reveal subtle information in the lineshape, such as **fine structure** (i.e. Kα12 in the Kα) or **satellite peaks** (i.e. Kβ' in the Kβ1,3 and Kβ'' in the Kβ2,5), high energy resolution is essential. When XAS and XES studies are performed using the so-called **wavelength-dispersive method**, a crystal or grating is placed to disperse the X-rays, enabling better energy discernment and improved resolution. High resolution is crucial for dynamically extracting information about the metal state, particularly in resolving the near-edge (XANES) and extended (NEXAFS) regions of the XAS edge and the XES emission lines. However, **incorporating these monochromators significantly complicates ray-tracing simulations of the setup**, necessitating more advanced modeling techniques to accurately capture their effects.

Furthermore, **there is a substantial gap in the integration of these methods to simulate the entire process, from ab initio calculations to the final spectrum**, due to the interdisciplinary expertise required across both theoretical and experimental domains.

> **OBJECTIVE 1 (O1):** I will integrate advanced **ab initio computational methods** with precise **ray-tracing simulations (WP2)** to comprehensively model the **K and L absorption edges** for **XAS**, and the **Kα, Kβ, and Lα emission lines** for **XES**. This integration aims to achieve a **prediction accuracy within 1%** when compared with experimental spectra.

Core-hole spectroscopies require the excitation of a core electron, achievable through various methods. Focusing specifically on X-ray approaches, absorption and emission studies necessitate an X-ray source. Traditionally, **synchrotrons have been the primary source for XAS/XES experiments**, as third- and fourth-generation synchrotron facilities provide highly intense, finely tunable beams precisely matched to the energies of interest [9].

However, **recent trends show increasing interest in laboratory-based, table-top X-ray setups** [2]. Such laboratory-based setups offer significant advantages, including greater availability and flexibility compared to synchrotron facilities, where access is both limited and highly competitive. Despite these benefits,

laboratory-based methods face significant challenges, most notably a drastically lower Signal-to-Background Ratio (SBR). This limitation arises primarily because X-ray tubes produce relatively lower intensities, with most of their power lost as heat [10].

To address this challenge, advanced optimization techniques are being developed to maximize the SBR for laboratory-based XAS and XES setups, significantly enhancing their sensitivity, particularly for operando experiments where real-time tracking of metal states is essential.

Machine Learning (ML) based methods, especially **Bayesian Optimization (BO)** and **Reinforcement Learning (RL)**, are increasingly employed to optimize these experimental conditions. Recent studies have demonstrated ML's effectiveness in maximizing SBR by optimizing parameters in laboratory-based X-ray setups. For instance, BO has successfully been applied to optimize data acquisition strategies for XAS [11] and to fine-tune X-rays beamline configurations for enhanced sensitivity and resolution [12].

Despite ongoing efforts, **there is still a lack of progress in creating a so-called digital twin, a copy of the experiment based on simulations that autonomously optimizes the setup components to maximize the SBR.**

> **OBJECTIVE 2 (O2):** I will establish and deploy an advanced **optimization framework employing ML techniques (WP3) trained on simulations obtained from WP2 methodologies**. Specifically, **I will use RL to iteratively refine setup parameters, aiming for a minimum tenfold improvement in the SBR for the target emission or absorption edge.** The effectiveness of the optimization approach will be rigorously validated by **comparison with conventional grid scan techniques.**

While most experimental setups have traditionally specialized exclusively in either XAS or XES, **recent advancements show an increasing interest in integrated systems capable of simultaneously performing both spectroscopy techniques**. Precise alignment to specific absorption edges or emission lines and careful fine-tuning to maximize the SBR are critical to optimizing the performance of both XAS and XES analyses. **Current integrated setups often utilize fixed positions or rely on partial external control**, underscoring a substantial gap in the availability of fully automated systems that can autonomously perform precise and adaptive positional adjustments and alignment.

> **OBJECTIVE 3 (O3):** I will design and build a **fully automated prototype setup for simultaneous XAS and XES**, with motorized control of all critical components, including the X-ray source, monochromator crystals, and detectors **(WP4)**. The automation will enable **precise alignment** with the specific energy regions of interest, **facilitate simultaneous XAS and XES measurements**, and optimize the fine movements of all components **to maximize the SBR**.

The **reduced SBR of laboratory-based setups compared to synchrotron sources** significantly limit their applicability, particularly for samples with low metal concentrations. While synchrotron-based XAS/XES measurements can be completed within seconds, laboratory-based methods typically require spectral acquisition times ranging from several minutes to hours. For example, the **Limit of Detection (LOD)** for XES has traditionally been in the **weight-percent (wt%) range of 0.1-1 wt% (i.e., 100-1000 ppm)** [2].

> **OBJECTIVE 4 (O4):** I will **apply the framework developed in WP4 to the automated and combined XAS/XES setup (WP5)**. This will include fine-tuning the setup components' positions and angles to maximize the SBR, achieving a **LOD below 0.01 wt% (< 100 ppm)**. The setup will maintain a **high energy resolution (E / ΔE > 4000)** for metals with edges/lines in the 5-20 keV range and complete measurements within an **acquisition time on the order of minutes**.

Recent advancements in laboratory-based techniques have significantly increased interest in performing operando measurements on metals [13]. Unlike ex-situ and in-situ methods, **operando approaches enable real-time monitoring of the spin and oxidation states of metals under operational conditions**, making them especially valuable in energy storage research. For instance, XAS and XES have been effectively utilized in lithium-ion batteries to observe changes in electronic states during battery operation. A notable example is LiFePO4, where XAS and XES clearly reveal the impact of the charging process **(see FIGURE C3)**.

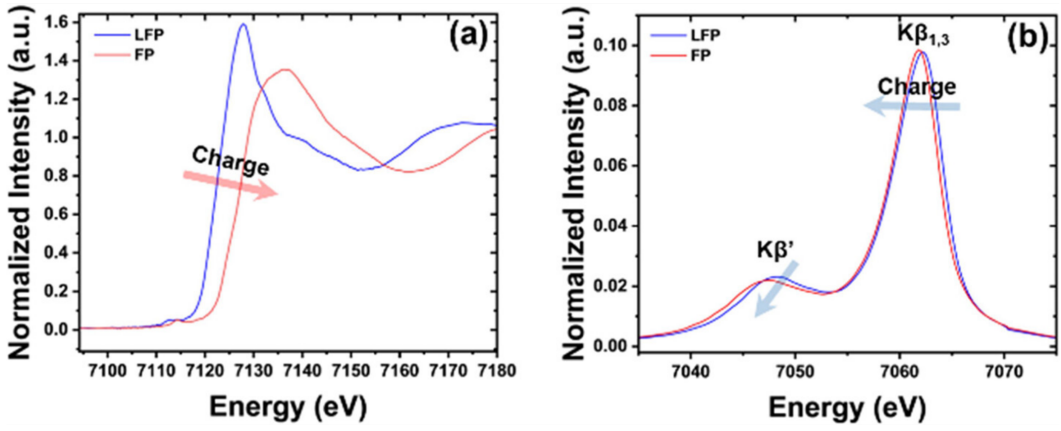


FIGURE C3: Charge effect in a $LiFePO_4$ battery on XAS and XES. Effect of charging on a battery in $LiFePO_4$ (LFP) on the XAS K-edge of iron (left) and on the XES Kβ line of iron, where effects on XAS are more pronounced compared to XES. Results taken from https://doi.org/10.1021/acsami.3c18424.

Quantitative analysis of X-ray emission spectra often relies on the **Integrated Absolute Difference (IAD)** method, which assesses spectral variations by comparing operando spectra to reference states. This approach has been successfully applied to **lithium-sulfur batteries**, where the **soft X-ray Kβ emission line** is used to track sulfur's electronic state variations [14]. However, the operando application of XAS and XES to track **transition-metal cathodes in lithium-ion batteries** remains limited, particularly in **laboratory-based setups**. A major challenge in the literature is the **difficulty of simultaneously applying XAS and XES** while ensuring sufficient **sensitivity and accuracy in probing oxidation and spin states**. Factors such as **X-ray penetration depth, electrode thickness, and cell design** significantly impact measurement reliability, yet these parameters are often **not optimized systematically**. Additionally, laboratory-based XAS/XES experiments

suffer from **long acquisition times and lower SBR** compared to synchrotron-based setups, limiting their applicability to **real-time, high-resolution operando measurements**.

> **OBJECTIVE 5 (O5):** I will finally validate the optimized XAS/XES setup through an **operando study of Li-ion battery cathodes**, focusing on **NMC ($LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$) and $LiFePO_4$**, whose **K-edges and Kα and Kβ emission lines** fall within the **5-20 keV range**. A **battery prototype** with **optimized cathode thickness** will be designed to enhance X-ray sensitivity for accurate **oxidation and spin state tracking**. The study will enable **real-time monitoring** during **charge/discharge cycles** with a **10-minute temporal resolution**.

In summary, the **Key Objectives of MOSES (see FIGURE C4) will narrow the gap between conducting X-ray spectroscopies in laboratory settings and performing them at synchrotron facilities**. This will be achieved through the combined use of simulation and optimization methods, supported by ML techniques, to enhance acquisition efficiency and sensitivity. **These improvements will expand the applicability of laboratory-based XAS and XES, enabling more frequent and accurate operando studies.**

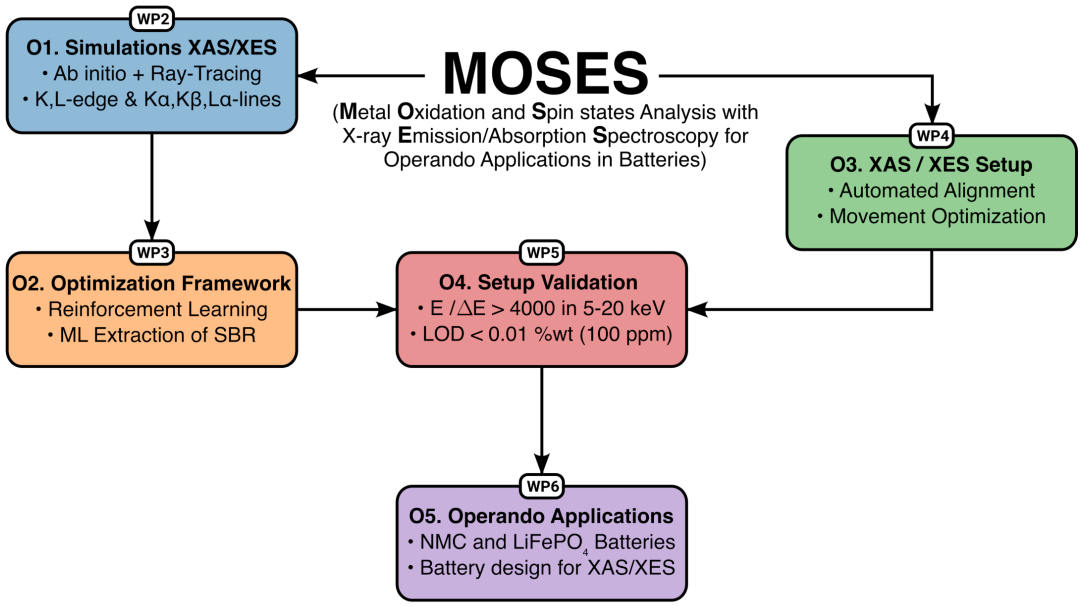


FIGURE C4: Key Objectives of MOSES. Schematic representation of the MOSES's key objectives, starting from the simulations for XAS and XES spectra (O1), the machine learning-driven optimization framework (O2), building the automated experimental setup (O3), the rigorous validation of the developed system (O4), and operando spectroscopy applications for real-time monitoring of metal states in battery research (O5). Each objective is related to the respective Work Package (WP).

REFERENCES

[1] Goodenough, J. B. Magnetism and the Chemical Bond, John Wiley 1963, I, 1− 385.

[2] https://doi.org/10.1016/j.ccr.2020.213466

[3] https://doi.org/10.1201/9781420008425

[4] https://doi.org/10.1017/CBO9780511809194

[5] https://doi.org/10.1088/0268-1242/29/2/023002

[6] https://doi.org/10.1016/j.ccr.2004.03.018

[7] https://doi.org/10.1021/acs.accounts.0c00171

[8] https://doi.org/10.1103/RevModPhys.75.35

[9] https://doi.org/10.1002/9783527697106

[10] https://doi.org/10.1259/bjr.1997.0002

[11] https://doi.org/10.1088/2632-2153/ad4253

[12] https://doi.org/10.1364/3D.2023.JW2A.32

[13] https://doi.org/10.1039/D1CP03263A

[14] https://doi.org/10.1021/acsaem.4c02330

Research Methodology:

Here, I outline the methodology of the **MOSES** project across its six **Work Packages (WPs)**. Project activities will be conducted at the **National Laboratory of Frascati (LNF) of INFN**. The core MOSES team will comprise myself, committed throughout the project's entire duration (60 months) and three postdoctoral researchers **(R1-R3)**, for a total 6 man-years. **WP1 is dedicated to personnel recruitment,** ensuring that team members are selected based on specialized skills and expertise, which are determined by the specific WP to which they are assigned.

WP2: AB INITIO AND RAY-TRACING SIMULATIONS (M1-M21)

In this WP, I will develop the computational methodologies for simulating the X-ray spectra needed for training the Machine Learning (ML) models of WP3. The activities will be carried out by myself and R1 over months M1 to M21. Initially, spectra will be computed using ab initio calculations to accurately represent intrinsic electronic properties **(see FIGURE C5)**. Subsequently, these calculated spectra will serve as inputs for detailed ray-tracing simulations, effectively incorporating experimental setup effects to produce realistic spectral data.

Ab initio Calculations

Absorption edges will be calculated using the software GPAW [1], a well-established computational tool employing **Density Functional Theory (DFT)** and beyond methods for first-principles simulations, extensively used in my previous research [2]. GPAW includes dedicated modules for X-ray Absorption Spectroscopy (XAS), allowing for precise determination of absorption edge lineshapes [3].

Emission lines can also be calculated using GPAW; however, for higher accuracy in transition metal complexes, correlation effects must be carefully included. Therefore, I will primarily employ the **Multi-Configurational Dirac-Fock** [4] method, previously used in my work within the VIP experiment [5], to accurately describe isolated metals. Ligand and crystalline environment effects will subsequently be introduced as perturbations, with their coupling strengths estimated from complementary DFT calculations using GPAW.

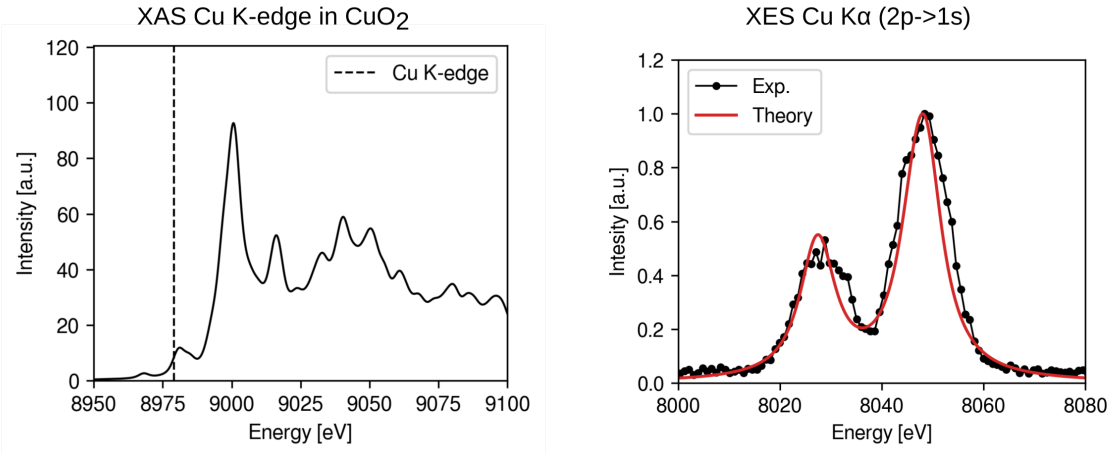


FIGURE C5: Ab initio Calculations for XAS and XES. Example calculations from first principles for the absorption K-edge for copper in CuO_2 (left) taken from Material Project (https://next-gen.materialsproject.org/materials/mp-14549?formula=CuO) with reference K-edge value with a dashed lines and on the right the Kα (2p->1s) emission line for an isolated copper atom (red) in comparison with the experimental XES spectrum (black dots) obtained with the VOXES setup at LNF.

Ray-Tracing simulations

I will perform ray-tracing simulations using a Monte Carlo approach with the **XrayTracer (XRT)** code [6] to accurately replicate the response function of the experimental setup. Additionally, these simulations will serve to validate the optimization methods developed in WP3 and help with the design of the setup in WP4.

WP3: DEVELOPING OPTIMIZATION FRAMEWORK (M19-M30)

The scope of this WP is to build an optimization framework that automatically moves and aligns the components of the XAS/XES setup **to maximize the Signal-to-Background Ratio (SBR) of the spectrum of the given absorption edge and emission line of interest**. The activity will be performed by me and R1 during M19-M30.

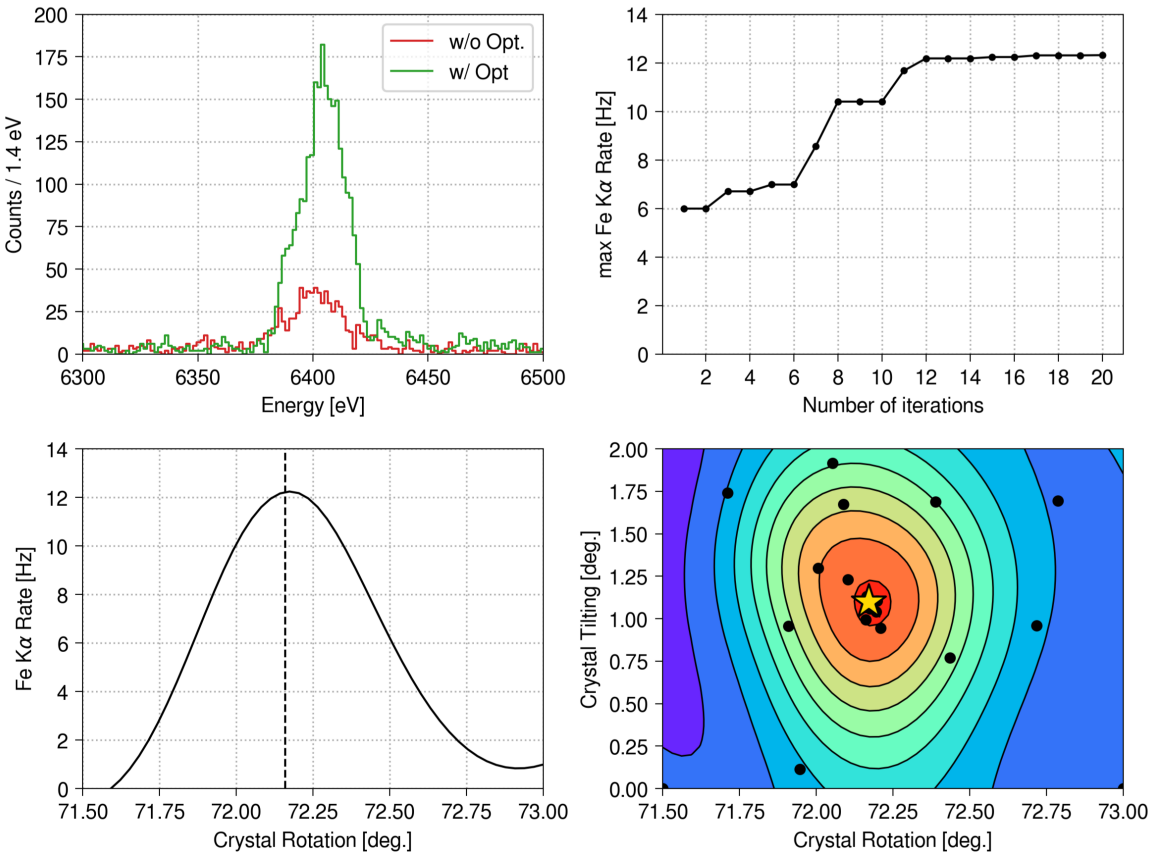


FIGURE C6: Bayesian Optimization of iron's XES Kα. The plot on (top left) displays the FeKα before and after the optimization of the rate, while on (top right) the maximum obtained rate is plotted as a function of the number of iterations, where for each iteration the rate is obtained from a fit of the spectrum. During the Bayesian optimization, the rotation and tilting of the crystal [see Figure (C9) of the setup] are adjusted to maximize the rate. The associated Gaussian Process is plotted as a function of the rotation in the (bottom left) plot, while the partial dependence plot for the two parameters is displayed in the (bottom left), with the maximum indicated with a star. The optimization process aims to find the optimal crystal rotation and tilting angles to maximize the Kα peak rate of iron. With a precision of 0.01° within the specified angle ranges, the optimization provides an improvement of over a factor of 10 compared to a simple grid search.

Optimization method

I have already explored the idea of optimizing SBR of XES spectra at the LNF, where, employing within the **VOXES** project [7], I aimed to optimize the SBR of the Kα line of iron by adjusting the angles of the monochromator crystal **(see FIGURE C6)**. The optimization method used there was **Bayesian Optimization (BO)**, where by fitting a **Gaussian Process (GP)** to the SBR as a function of the crystal angles, it iteratively proposed points to sample to maximize the SBR. However, **this approach has the following limitations: it scales poorly with the number of parameters due to the use of GP to fit the SBR, and it relies on extrapolating the SBR with a fit from the spectra.**

For the MOSES project, I aim to solve the first issue by adopting a **Reinforcement Learning (RL)-based optimization framework [8]**. Unlike **BO**, which can become **computationally expensive in high-dimensional parameter spaces**, RL can **learn an efficient policy** through interaction with a predictive model of the system. Specifically, I will implement a **model-based RL approach**, where the **spectral models developed in WP2 serve as a surrogate environment** within the RL framework. This allows the RL agent to **explore and optimize the parameter space efficiently** while making use of **prior knowledge embedded in the spectral models**. Compared to BO, this approach is **expected to scale better** with increasing parameter dimensions, **provided that the model-based predictions accurately reflect the system's behavior**.

Machine Learning Models

To extract the spectra's SBR without relying on spectrum-specific fitting, I will employ ML techniques. The training strategy and architecture of the ML model are illustrated in **FIGURE C7**. Initially, I will generate a training dataset by simulating XAS/XES spectra using methods developed in WP2. A Conditional Variational Autoencoder (CVAE) [7] will be employed, supervised by explicitly providing the target SBR as a conditional vector to the decoder. This approach enables the encoder to directly estimate SBR from spectral input, facilitating real-time optimization via RL. **My prior experience with similar machine learning approaches, albeit in different contexts, will be valuable and transferable** [9].

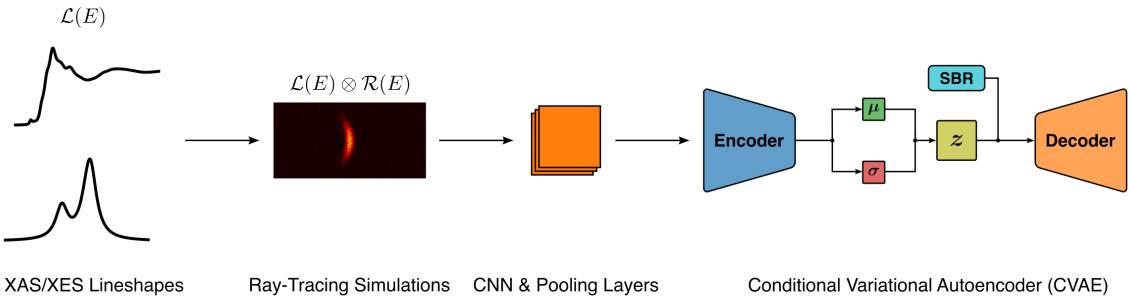


FIGURE C7: ML Model for SBR Extraction. The Machine Learning (ML) model designed to extract the Signal-to-Background Ratio (SBR) from spectra. Initially, XAS/XES spectra are simulated using ab initio methods to generate accurate lineshapes $\mathcal{L}(E)$. These spectra are then convolved with the detector response function $\mathcal{R}(E)$, obtained through ray-tracing simulations, to produce realistic detector outputs. The resulting dataset is used to train ML models on varying SBR values, particularly employing a Conditional Variational Autoencoder (CVAE), enhanced with convolutional and pooling layers, enabling the generation of spectra with specified SBR characteristics.

The workflow of the optimization framework is depicted in FIGURE C8. There is an ML sub-workflow involving the training of ML models on the simulations developed in WP2. Subsequently, the optimization loop controls the movements of the setup to iteratively find the optimal positions of the various components.

WP4: BUILDING XES / XAS SETUP (M1-M27)

In this WP, the prototype setup for combined XAS and XES will be constructed. R2 and myself will participate throughout the first three years of the project (M1–M27).

XAS/XES Prototype Setup

- Fully integrated **system combining XAS and XES** capabilities.
- **Automated alignment** targeting optimal absorption edges and emission lines.
- Motorized stages for **fine positioning**, aimed at maximizing the SBR.

Components (see FIGURE C9)

- **X-ray tube source** capable of energies between 5–20 keV, requiring a maximum voltage of 60 kV and current of at least 50 mA (critical for maximizing SBR).
- **Two mosaic crystals** with motorized positioning for precise energy dispersion.
- **Two pixel detectors** optimized for capturing 2D images, providing comprehensive data for training the ML models.
- **Slit systems** for beam shaping and reduction of background.
- **Motorized sample holder** for automated and reproducible positioning.

Pixel detectors will be utilized to capture 2D images, essential for correcting significant dispersion effects perpendicular to the propagation direction, and providing richer datasets for training ML algorithms, thereby enhancing SBR extraction accuracy.

WP5: MEASUREMENTS VALIDATION (M28-M36)

This WP will apply the computational framework developed in WP3 to the fully automated prototype setup built in WP4. The activity will be led by **R2 under my supervision**, focusing on validating the setup and developing custom **control software** for operation of hardware components and data acquisition (DAQ).

Benchmark measurements will be conducted using **standard metal foils (Fe, Co, Ni, Cu) and transition metal compounds (Mn, Cr)** to establish reference K/L edges and Kα and Kβ lines within the **5-20 keV range**. These materials will be **cross-validated against synchrotron data** to ensure accuracy. Initial alignments will employ **manual grid scans**, optimizing key parameters such as **monochromator angles and detectors height** to enhance **SBR** and establish a **performance baseline**.

The **automated optimization framework** will then refine up to **30 experimental parameters** to surpass traditional grid scanning. Performance will be evaluated based on the **SBR improvement (target factor of more than 10 over manual scans).** The system's energy resolution will be tested using **high-purity metallic foils**, targeting **E/ΔE = 4000,** sufficient for resolving oxidation state shifts (0.1-0.5 eV) and spin state variations (0.5-1 eV) in transition metal cathodes (e.g., $Fe^{3} \rightarrow Fe^{2}$ in $LiFePO_4$, $Ni^{2} \rightarrow Ni^{3}$ in NMC). The **LOD will be rigorously validated** using dilution series of metal solutions, defining LOD as the **signal level 3σ above background noise**.

By validating energy resolution, sensitivity, and stability, **WP5 will establish the laboratory-based XAS/XES setup as a reliable alternative to synchrotron spectroscopy** for operando applications.

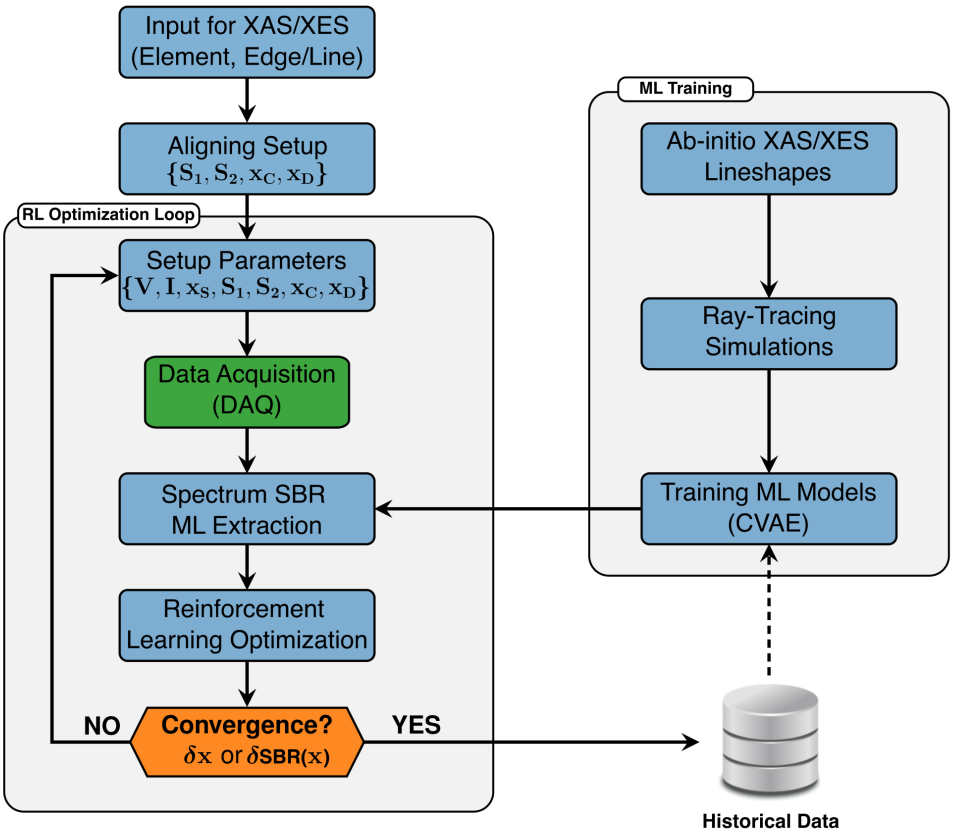


FIGURE C8: MOSES Optimization Workflow. The optimization workflow involves several key steps. First, the framework aligns the setup based on the selected input absorption edge or emission line and initiates the measurement with the DAQ, where x_C and x_D are collective coordinates for crystals and detectors [see FIGURE C9 of the setup]. The Signal-to-Background Ratio extraction is done using Machine Learning (CVAE), previously trained on the ray-tracing simulations. The optimization loop continues until convergence, depending on the threshold over the parameters x or on the SBR, then the results are saved in a database, that eventually can be used for training the CVAE on historical real data.

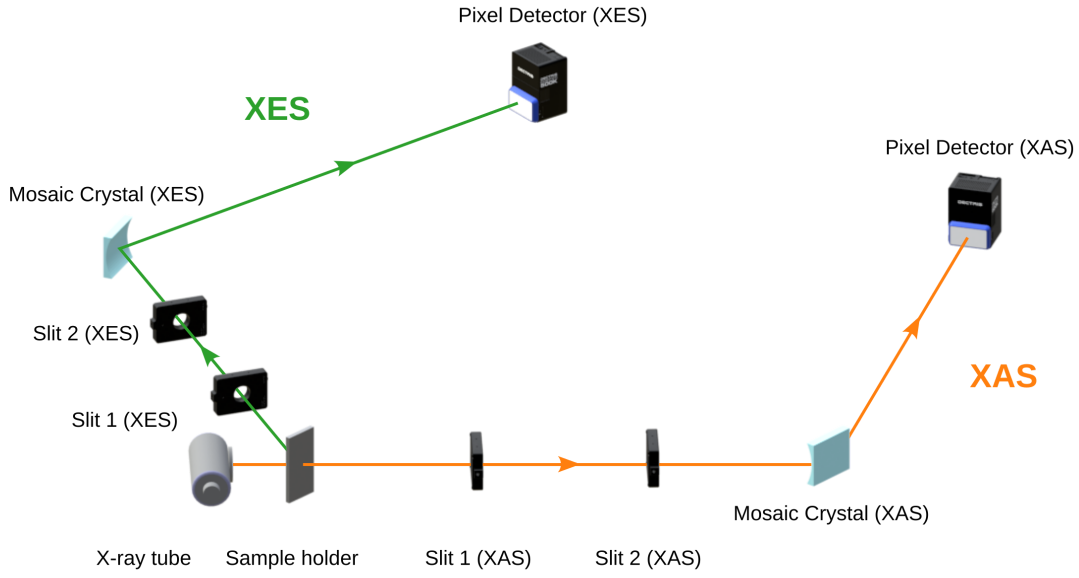


FIGURE C9: Prototype Setup for combined XAS and XES. The proposed setup for the MOSES project comprises an X-ray tube, a sample holder, and two lines: one for XAS (orange) and one for XES (green). Each line includes two slits, a mosaic crystal, and a pixel detector. Mosaic crystals and pixel detectors will have motorized stages for aligning to the edge/line of interest. Addiotionaly, together with the sample holder, they will also have axis for fine aligment to optimize the signal-to-background ratio of the spectra.

WP6: OPERANDO APPLICATIONS & DISSEMINATION (M37-60)

This WP will demonstrate the capabilities of the optimized XAS/XES setup through operando applications, specifically focusing on Lithium-ion batteries, targeting high-resolution spectroscopy of lithium-ion cathodes such as **NMC ($LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$)** and **$LiFePO_4$.**

A **thin-film cathode** (~10-20 µm) will be fabricated to optimize X-ray penetration. A **Kapton** or **Beryllium window** will be utilized, carefully selected to balance transmission efficiency with structural integrity. Additionally, **the electrolyte (1M LiPF in EC/DMC), separator, and overall cell architecture, including graphite anode, copper current collector, and aluminum current collectors, will be meticulously optimized to minimize X-ray attenuation**, thereby maximizing signal quality while preserving battery performance and structural stability. Detailed simulations of X-ray absorption will guide the precise determination of the optimal thickness for each component, maximizing sensitivity and ensuring accurate operando measurements. **An illustrative concept of the battery cell is depicted in FIGURE C10**.

Initial experiments will involve in-situ XAS/XES analyses on fully charged and discharged states to establish benchmark spectral shifts. Subsequently, operando measurements will track spectral evolution during battery cycling, achieving a temporal resolution of seconds, enabling the observation of transient phenomena.

The key performance indicators include:

- Precise tracking of **oxidation states** (e.g., Fe^{3} $\rightarrow Fe^{2}$) with an energy precision ≤0.1 eV.
- **Spin-state analysis** via XES using the **Integrated Absolute Difference** (IAD) method.
- Benchmarking results against synchrotron-derived data to ensure reproducibility and validate operando measurements.

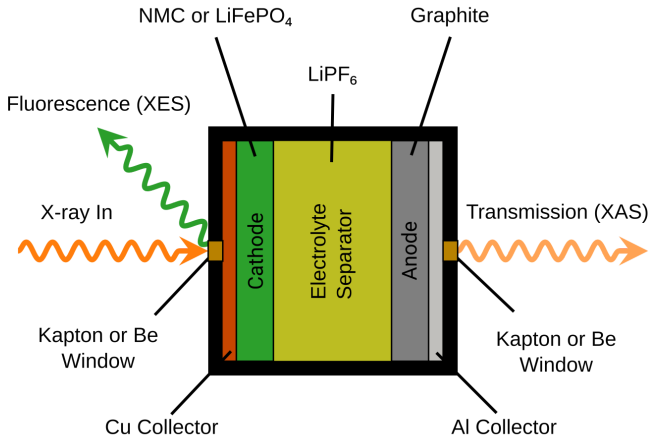


FIGURE C10: Operando Battery Cell Design. The schematic illustrates an operando battery cell designed for simultaneous XAS and XES measurements. Incident X-rays (orange) pass through a Kapton/Beryllium entrance window and interact with the battery materials (Anode, Electrolyte Separator, Cathode). Transmitted X-rays (orange, right) are collected for X-ray Absorption Spectroscopy (XAS), while fluorescence X-rays (green, top-right) are emitted from the cathode and detected for X-ray Emission Spectroscopy (XES). This setup enables real-time tracking of oxidation state and spin state changes in battery materials during charge/discharge cycles.

I expect that the MOSES will result in approximately 10 scientific papers published in international peer-reviewed journals with open access and shared datasets to support reproducibility. Project staff will participate in conferences and workshops, and there will be an opportunity to transport the setup to a synchrotron facility to validate the operando measurements. Additionally, a dedicated **final workshop will be organized** to disseminate project results and foster engagement with the X-ray spectroscopy and battery research communities.

The project's M**ilestones (MS)**, closely aligned with each Work Package, are outlined below along with their expected completion months (M):

- **MS1: Delivery of Pixel Detectors (M12)**
- **MS2: Implementation of Optimization Framework (M30)**
- **MS3: Completion of Experimental Setup (M36)**
- **MS4: Submission of Operando Measurements Paper (M57)**
- **MS5: Hosting of Final Workshop (M60)**

The **Gantt chart** summing up all Work Packges, Milestones, Taks and personnel involvement is reported in **FIGURE C11.**

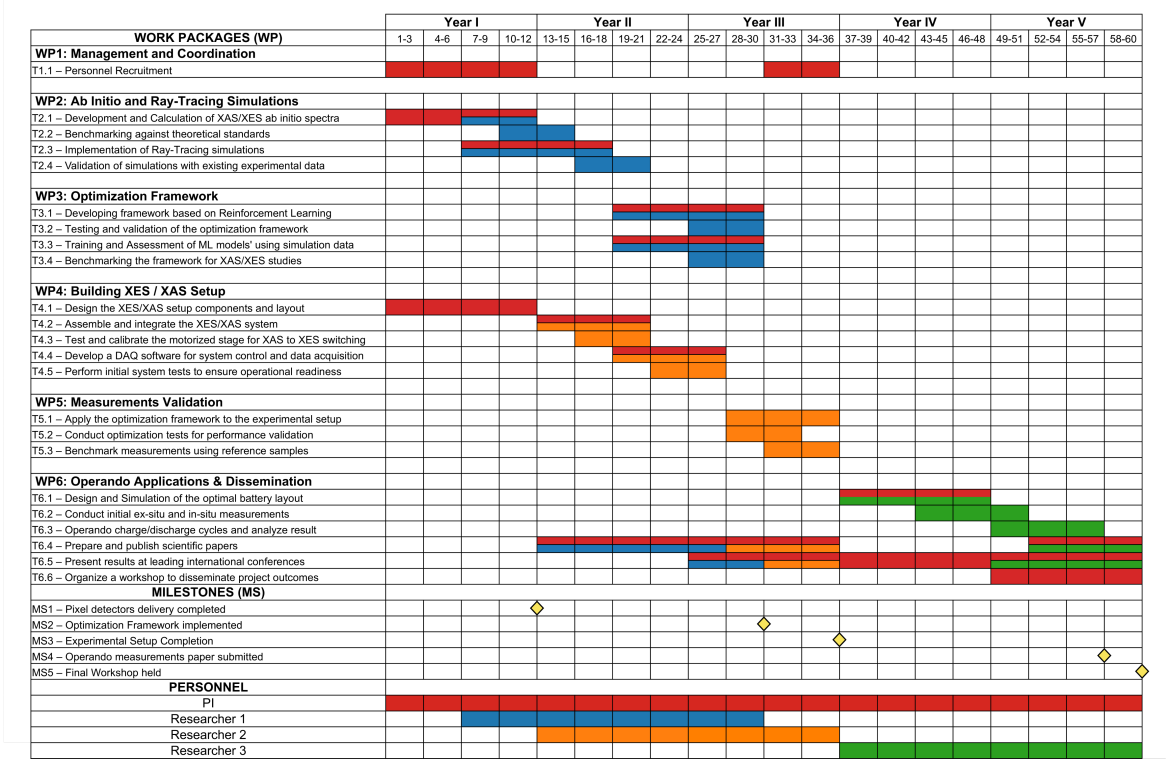


FIGURE C11: Gantt chart of the MOSES project. This Gantt chart illustrates the timeline of the MOSES project, spanning five years and organized by Work Package (WP) and Task (T). Project Milestones (MS) are indicated along with their expected release dates. The Gantt chart also includes personnel involvement, represented with individual colors for each team member. Tasks involving multiple contributors are highlighted using a combination of colors to reflect collaboration.

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PROJECT SCHEDULE

Totale Milestone: 1.225.544,00

N°: 1	Milestone: Pixel detectors delivery completed	Total Cost: 114.060,00

N°	Activity	Release Month (n°)	Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6	Semester 7	Semester 8	Semester 9	Semester 10	Total cost

N°	Activity	Release Month (n°)	Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6	Semester 7	Semester 8	Semester 9	Semester 10	Total cost
1	Pixel detectors delivery completed	12	6	6	0	0	0	0	0	0	0	0	114.060,00

N°: 2	Milestone: Optimization Framework implemented	Total Cost: 228.946,00

N°	Activity	Release Month (n°)	Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6	Semester 7	Semester 8	Semester 9	Semester 10	Total cost
1	T1.1 - Personnel recruitment	6	6	0	0	0	0	0	0	0	0	0	55.728,00
2	T2.1: Development and calculation of XAS/XES spectra	12	6	6	0	0	0	0	0	0	0	0	31.810,00
3	T2.2: Benchmarking against theoretical standards	15	0	3	3	0	0	0	0	0	0	0	10.570,00
4	T2.3: Implementation of Ray-Tracing simulations	18	0	6	6	0	0	0	0	0	0	0	41.830,00
5	T2.4: Validation of simulations with existing experimental data	21	0	0	3	3	0	0	0	0	0	0	9.280,00
6	T3.1: Developing framework based on Reinforcement Learning	30	0	0	0	6	6	0	0	0	0	0	32.788,00
7	T3.2: Testing and validation of the optimization framework	30	0	0	0	0	6	0	0	0	0	0	7.076,00
8	T3.3: Training and Assessment of ML models' using simulation data	30	0	0	0	6	6	0	0	0	0	0	32.788,00
9	T3.4: Benchmarking the framework for XAS/XES studies	30	0	0	0	0	6	0	0	0	0	0	7.076,00

N°: 3	Milestone: Experimental Setup Completion	Total Cost: 381.374,00

N°	Activity	Release Month (n°)	Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6	Semester 7	Semester 8	Semester 9	Semester 10	Total cost
1	T1.1 - Personnel recruitment	12	0	6	0	0	0	0	0	0	0	0	53.148,00
2	T4.2: Assemble and integrate the XES/XAS system	21	0	0	6	3	0	0	0	0	0	0	85.338,00
3	T4.3: Test and calibrate the motorized stage for XAS to XES switching	21	0	0	3	3	0	0	0	0	0	0	32.690,00
4	T4.4: Develop a DAQ software for system control and data acquisition	27	0	0	0	6	3	0	0	0	0	0	62.369,00
5	T4.5: Perform initial system tests to ensure operational readiness	27	0	0	0	3	3	0	0	0	0	0	39.230,00
6	T5.1: Apply the optimization framework to the experimental setup	36	0	0	0	0	3	6	0	0	0	0	48.236,00
7	T5.2: Conduct optimization tests for performance validation	33	0	0	0	0	3	3	0	0	0	0	30.826,00
8	T5.3: Benchmark measurements using reference samples	36	0	0	0	0	0	6	0	0	0	0	29.537,00

N°: 4	Milestone: Operando measurements done	Total Cost: 457.944,00

N°	Activity	Release Month (n°)	Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6	Semester 7	Semester 8	Semester 9	Semester 10	Total cost
1	T1.1 - Personnel recruitment	36	0	0	0	0	0	6	0	0	0	0	55.728,00
2	T6.1: Design and finalize the battery layout	48	0	0	0	0	0	0	6	6	0	0	77.400,00
3	T6.2: Conduct initial in-situ measurements	51	0	0	0	0	0	0	0	6	3	0	20.640,00

N°	Activity	Release Month (n°)	Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6	Semester 7	Semester 8	Semester 9	Semester 10	Total cost
4	T6.3: Operando charge/discharge cycles and analyze result	57	0	0	0	0	0	0	0	0	6	3	15.480,00
5	T6.4: Prepare and publish scientific papers	60	0	0	6	6	6	6	0	0	3	6	117.168,00
6	T6.5: Present results at leading international conferences	60	0	0	0	0	6	6	6	6	6	6	171.528,00

N°: 5	Milestone: Final Workshop held	Total Cost: 43.220,00

N°	Activity	Release Month (n°)	Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6	Semester 7	Semester 8	Semester 9	Semester 10	Total cost
1	T6.6: Organize a workshop to disseminate project outcomes	60	0	0	0	0	0	0	0	0	6	6	43.220,00

GANTT CHART

GANTT_MOSES.pdf

EXPECTED IMPACT

The approach proposed by the MOSES project to study metal states under operando conditions using X-rays is expected to have a significant impact across multiple sectors **(see FIGURE C12)**.

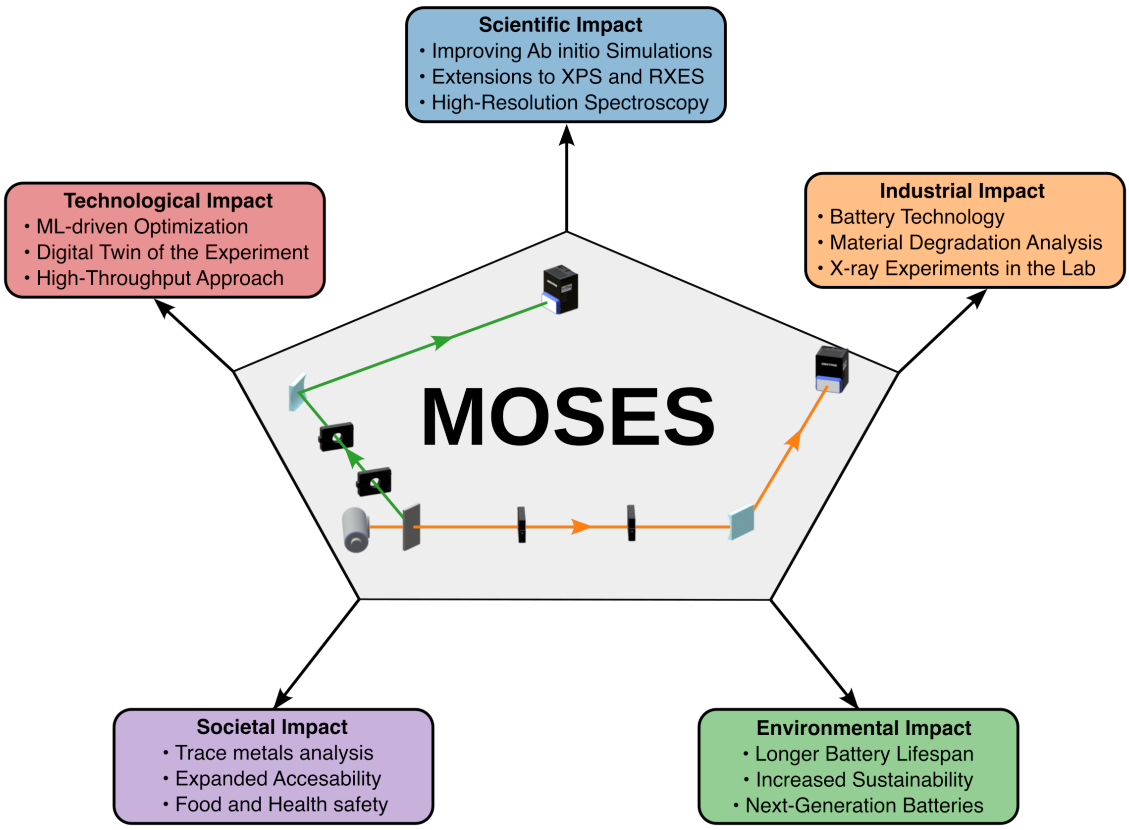


FIGURE C12: MOSES Project Impact Overview. The MOSES project advances X-ray spectroscopy with impacts in science, technology, industry, society and environment.

Scientific Impact:

- The combination of **ab initio methods and ray-tracing simulations** will enhance the modeling of XAS and XES spectra, directly correlating theoretical parameters with experimental results. This will improve **the accuracy of spectral predictions**, enabling deeper insights into the **electronic structure and dynamic behavior** of materials in real-time conditions. The **integration of first-principles calculations** with experimental validation will also advance the development of novel spectral analysis techniques
- Beyond XAS and XES, the methodologies developed within MOSES can be extended to **X-ray Photoelectron Spectroscopy (XPS) and Resonant XES (RXES)**, broadening the scope of multi-modal spectroscopic approaches. This will enable **correlation studies between different techniques**, allowing researchers to derive a more comprehensive understanding of **metal oxidation, spin states, and ligand interactions** in complex materials.
- The optimized high-resolution X-ray spectroscopy setup developed in MOSES will also be applicable in detecting X-rays in **particle physics**, particularly for the study of **exotic atoms** such as **kaonic atoms at DAFNE and J-PARC, and pionic and muonic atoms at PSI**. Additionally, the improved spectral resolution and precision will enable experiment where X-rays are used as stringent tests of **quantum mechanics**, including investigations of the **Pauli exclusion principle (VIP experiment) and the Generalized Uncertainty Principle (GUP).**

Technological Impact:

- MOSES will pioneer the use of **machine learning-driven optimization** to develop a **digital twin** of the X-ray spectroscopy experiment. This will enable real-time adaptation of experimental parameters, **reducing manual intervention** and significantly improving **efficiency, reproducibility, and automation** in X-ray spectroscopy.
- The automation process for combined XAS and XES measurements will enable simultaneous data acquisition, **bringing a high-throughput approach to X-ray spectroscopies** and reducing experimental costs while increasing reproducibility. Automation will also allow for the rapid screening of materials, expediting the development cycle.

Industrial Impact:

- MOSES has strong industrial relevance, particularly in sectors such as **battery technology, materials science, and catalysis**. The development of an optimized industrial workflows it will provide predictive tools to test **material degradation and quality control**. This capability will be critical for **battery**

manufacturers, semiconductor industries, and materials research laboratories.

- The automation of **XAS/XES characterization** will allow companies to **streamline production processes, improve operational efficiency, and reduce experimental costs**. This will make advanced spectroscopic techniques more accessible to industries that previously relied on **synchrotron facilities**, thus promoting the adoption of **lab-based high-resolution X-ray analysis** in industrial R&D.

Societal Impact:

- MOSES will advance the **limit of detection** in X-ray analysis, extending its applications to **biological, environmental, and food analysis**. The ability to detect **trace metal concentrations** and study **chemical state variations** will contribute to improved **quality control, food safety, and environmental monitoring**, benefiting both **regulatory bodies and public health sectors.**
- By improving **accessibility to high-resolution X-ray spectroscopy**, MOSES will enable **smaller research institutions and industries** to perform advanced materials characterization **without the need for synchrotron access**. This will democratize **cutting-edge X-ray spectroscopy**, fostering a broader adoption of advanced analytical techniques.
- The reduction in **measurement time and human intervention** will minimize exposure to **ionizing radiation**, improving workplace **safety for researchers and operators**. The automation framework will help **mitigate health risks** associated with **long-duration X-ray exposure**, aligning with **EU radiation protection directives** and international safety regulations.

Environmental Impact:

- The **application of operando X-ray spectroscopy in battery research** will enable the study of **cathode degradation during charge/discharge cycles**, leading to the development of more **sustainable and long-lasting materials** for energy storage applications. This aligns with the global push toward **sustainable energy solutions** and the **EU Green Deal objectives** for clean energy transitions
- By enhancing **battery lifetime and recyclability**, MOSES will contribute to reducing the **carbon footprint** of battery production. The insights gained from these studies will **accelerate the development of next-generation batteries**, supporting innovations in **electric mobility and grid-scale energy storage**

In summary, by integrating advanced computational simulations, machine learning-driven optimization, and fully automated experimental techniques, MOSES is well-positioned to improve operando X-ray spectroscopy and support meaningful progress in energy, industry, and society toward a more sustainable future.

HUMAN RESOURCES

Principal Investigator

Time, in percentage terms dedicated to research activities: 100

Time, in percentage terms dedicated for other activities of the project (e.g. management, and administrative activities) (for PI ≥ 5%): 20

	N° of units involved in the project:	Time Commitment (months/person):
Principal-Investigator:	1	60
Senior Staff:	0	0
Post Docs:	3	72
PHD Student:	0	0
Other Support Staff:	0	0
Total:	4	132

Detailed description of the research activities:
Within MOSES, three dedicated researchers (R1, R2, and R3) will be recruited at different stages of the project, each contributing unique expertise to ensure a well-balanced synergy between theory, simulations, experiment, and applications. All researchers will be employed at the LNF of INFN, with clear and complementary roles.

• Researcher 1 (R1) – Simulation & Machine Learning Specialist

-Recruitment: Months 1-7
-Duration: 24 months (Months 7-30)
-Expertise required: Computational physics, ab initio simulations, ray-tracing, machine learning
R1 will be selected within the first semester of the project based on expertise in ab initio simulations (e.g., DFT-based methods for X-ray spectroscopy), ray-tracing, and Machine Learning (ML) applications relevant to the optimization of the experimental setup. This researcher will work on WP1 and contribute significantly to WP2.
-Year I Responsibilities: Develop and optimize computational tools for ab initio simulations of XAS and XES, implementing methodologies for simulating absorption edges and emission lines of different metal oxidation and spin states. Additionally, initiate ray-tracing simulations to evaluate setup efficiency.
-Year II Responsibilities: Refine the ray-tracing models with experimental data and develop the optimization framework of WP2 integrating ML-based algorithms for experimental setup alignment.
Outcome Contribution: R1 will play a crucial role in the first scientific publication of the project, validating theoretical predictions and simulation frameworks before experimental implementation.

• Researcher 2 (R2) – Experimental X-ray Spectroscopy Specialist

-Recruitment: Months 7-12
-Duration: 24 months (Years III-IV, Months 13-36)
-Expertise required: X-ray spectroscopy, instrumentation, experimental physics, optics
-R2 will be selected in the second half of Year 1, focusing on experimental skills in X-ray spectroscopy with prior experience in XES and XAS techniques. This researcher will be instrumental in WP3, which involves designing, assembling, and testing the experimental setup for combined XAS and XES.
-Year III Responsibilities: Design and optimize the experimental setup, integrating necessary components such as X-ray sources, monochromators, and detectors. Develop calibration and alignment protocols to ensure reliable spectroscopic measurements.
-Year IV Responsibilities: Perform measurements, validate theoretical and ray-tracing predictions with experimental data, and contribute to data interpretation and analysis.
-Outcome Contribution: R2 will be actively involved in paper writing after successful validation of the experimental setup.

• Researcher 3 (R3) – Applied Chemistry & Functional Materials Expert

-Recruitment: Months 31-36
-Duration: 24 months (Months 37-60)
-Expertise required: Chemistry, electrochemistry, energy materials, batteries
The final researcher, R3, will be recruited at the end of Year 3, focusing on chemical applications of X-ray spectroscopy, particularly for battery and catalysis studies. Her/his work will primarily contribute to WP6, ensuring that spectroscopic methodologies are effectively applied to study metal cathodes in Li-ion batteries and catalytic materials under operational conditions.
Year IV Responsibilities: Design the battery layout and perform first insitu tests on the setup.
-Year V Responsibilities: Lead in-situ and operando studies, analyze degradation mechanisms in battery cathodes, and explore chemical reactivity in catalytic systems. Facilitate the transition from fundamental spectroscopy to applied research in energy materials.
-Outcome Contribution: R3 will be deeply involved in the final publications of the project, participate in conference presentations and workshops, and contribute to the final dissemination event.

To ensure continuity and knowledge retention, R1 and R2 will overlap for at least 12 months, facilitating the transition from simulation to real setup. This overlap is crucial for skill transfer in experimental techniques, setup maintenance, and data analysis methods, ensuring that expertise built in the computational phase is effectively applied in the experimental implementation.

ETHICAL ASPECTS

Indication and explanation of relevant ethical issues of the project::

There are no relevant ethical issues to report.

BUDGET PLAN

Expanditure item

Expenditure item	Year 1	Year 2	Year 3	Year 4	Year 5	Total
A. Principal Investigator	61.920,00	61.920,00	61.920,00	61.920,00	61.920,00	309.600,00
B. Personal Costs	30.960,00	123.840,00	92.880,00	61.920,00	61.920,00	371.520,00
Senior Staff	0,00	0,00	0,00	0,00	0,00	0,00
Post Doc	30.960,00	123.840,00	92.880,00	61.920,00	61.920,00	371.520,00
PHD Students	0,00	0,00	0,00	0,00	0,00	0,00
Other Support Staff	0,00	0,00	0,00	0,00	0,00	0,00
C. Instruments and equipment	96.000,00	0,00	0,00	0,00	0,00	96.000,00
D. Other Operational Costs	10.000,00	196.333,00	41.358,00	19.753,00	44.756,00	312.200,00
E. Scientific and technical consulting	0,00	0,00	0,00	0,00	0,00	0,00
Total not including indirect costs (€)						1.089.320,00
F. Overheads (€)						136.224,00
Total (€)						1.225.544,00
Additional Contribution						12.256,00
Total including Additional Contribution						1.237.800,00

Budget Description For Each Item In The Table :

Here I will report details of the cost reported in the table.

A. Principal Investigator (€309,600)
The budget allocated for the Principal Investigator (PI) is essential for the successful execution of the project. The PI will be responsible for the overall management, scientific direction, and coordination of the research activities. This includes supervising the experimental setup, ensuring the implementation of machine learning models for optimizing the setup parameters, and overseeing the training of models on ab initio simulations. Additionally, the PI will play a crucial role in disseminating results through publications and conference presentations.
The allocated budget of €309,600, divided into yearly salaries of €61,920 over the five years, ensures the PI's full-time commitment (100%) to the project. The PI's duties also include preparing progress reports, and ensuring compliance with the project's objectives and milestones. Furthermore, the PI will mentor postdoctoral researchers and collaborate with external partners to enhance the project's impact. The salary allocation is consistent with those indicated by the host institution, LNF of INFN.

B. Personnel Costs (€371,520)
Personnel costs for three post-doctoral researchers (totaling six man-years) represent a significant portion of the project's budget. No PhD students or additional support staff are included. The total cost allocated for post-doctoral researchers amounts to €371,520, with an individual annual cost of €61,920 over the entire six man-years of the project. The cost variations across different years are due to overlapping employment periods, particularly during the fourth year when the second and third post-doc positions coincide.

C. Instruments and equipment (€96,000)
The instrument costs refer to the customized pixel detectors (Eiger2 by Dectris) for the two lines (XAS+XES) of the proposed prototype setup. Each detector costs €60,000. In the budget, only the portion eligible for amortization under INFN's 60-month amortization policy is reported. Since the detectors will be delivered in the first year, the remaining 12 months of amortization will be covered by the KAONNIS and VIP experiments of INFN at LNF.

D. Other Operational Costs (€312,200)
The operational cost for developing this prototype for operando XAS and XES measurements amounts to €312,200. Costs shared among different years are publications (including Open Access fees), etc. over the year III-V of a total of €15,000 for ten papers (around €1,500 per paper) and cost per travels shared between

years III to V.
The cost allocated across the years:

• Year I (€10,000)

-Workstation (€10,000) required to perform the ab initio and ray-tracing simulations.

• Year II (€196,333)

-Portable GPU (Jetson Xavier by NVIDIA €3,000) to be associated with the setup.

-X-ray tube (€15,000)

-Sample holder motorize stages and support (€12,000): with four axes at €3,000 per axis

-Slits configuration (€15,000): including five systems (four operational and one spare), each costing €3k

-Mosaic crystals (€60,000): For the XAS and XES line, each one with three different curvature radius and two values of the mosaicity with an estimated cost of €5,000 per unit.

-Detector motorized stages and support (€72,000): For the XAS and XES line, for each crystal and detector pairs with six axes at €3,000 per axis

-Stages for alignment plus support structure (€16,000): For large movements to the specific edge/line of interest

• Year III (€41,358)

-Target samples for measurement validation.

• Year IV (€19,753)

• Year V (€44,756)

-Workshop.

Total Budget (€1,249,544)

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Work Package (WP), Task (T) and Milestone (MS)

WP1: Management and coordination

• T1.1 - Personnel recruitment (M1-M6,M31-M36)

WP2: Ab Initio and Ray-Tracing Simulations

• T2.1 - Development and calculation of XAS/XES spectra (M1-M12)

• T2.2 - Benchmarking against theoretical standards (M10-M15)

• T2.3 - Implementation of Ray-Tracing simulations (M7-M18)

• T2.4 - Validation of simulations with existing experimental data (M16-M21)

WP3: Optimization Framework

• T3.1 - Developing framework based on Reinforcement Learning (M19-M30)

• T3.2 - Testing and validation of the optimization framework (M25-M30)

• T3.3 - Training and Assessment of ML models using simulation data (M19-M30)

• T3.4 - Benchmarking the framework for XAS/XES studies (M25-M30)

WP4: Building XES / XAS Setup

• T4.1 - Design the XES/XAS setup components and layout (M1-M12)

• T4.2 - Assemble and integrate the XES/XAS setup (M13-M21)

• T4.3 - Test and calibrate the motorized stage for XAS to XES switching (M16-M21)

• T4.4 - Develop a DAQ software for system control and data acquisition (M19-M27)

• T4.5 - Perform initial system tests to ensure operational readiness (M22-M27)

WP5: Measurements Validation

• T5.1 - Apply the optimization framework to the experimental setup (M28-M36)

• T5.2 - Conduct optimization tests for performance validation (M28-M33)

• T5.3 - Benchmark measurements using reference samples (M31-M36)

WP6: Operando Applications & Dissemination

• T6.1 - Design and finalize the battery layout (M37-M48)

• T6.2 - Conduct initial in-situ measurements (M43-M51)

• T6.3 - Operando charge/discharge cycles and analyze result (M49-M57)

• T6.4 - Prepare and publish scientific papers (M13-M36,M49-M60)

• T6.5 – Travels and Presenting results at international conferences (M25-M60)

• T6.6 - Organize a workshop to present project outcomes (M49-M60)

MILESTONES:

MS1: Pixel detectors delivery completed (M12)

MS2: Optimization Framework implemented (M30)

MS3: Experimental Setup Completion (M36)

MS4: Operando Measurements Paper Submitted (M57)

MS5: Final Workshop held (M60)

Risk assessment and mitigation

• Risk 1: Discrepancy between theoretical predictions and simulation outcomes in the optimization framework.
Mitigation 1: Incorporate historical experimental data and generate additional training datasets using a CVAE to refine predictive accuracy and reduce deviations.

• Risk 2: The desired limit of detection (LOD) remains unattainable despite optimization efforts.
Mitigation 2: Utilize a high-power X-ray tube with a current of up to 100 mA and implement larger-area, thicker mosaic crystals to enhance signal intensity, carefully balancing energy resolution trade-offs.

• Risk 3: Challenges in conducting operando measurements on a functioning battery due to signal limitations.
Mitigation 3: Enhance sensitivity by implementing strategies from Mitigation 2. If challenges persist, perform XAS and XES separately, designing specialized cells tailored for each X-ray line to maximize measurement reliability.

DECLARATION AND DOCUMENTATION

Ethics self-assessment:

Allegato-5_Template_relativo_agli_aspetti_etici.pdf

Commitment Letter:

Commitment_Letter_INFN_FIS3_Manti.pdf

Is the commitment letter signed by the legal representative of the Host institution ?

YES

Delegation of the legal representative of the host institution:

I hereby declare I have read the information on the personal data processing policy, pursuant to EU Regulation 2016/679 and Legislative Decree no. 196/2003 and subsequent amendments for the purposes of managing this procedure, as well as to comply with the law within the scope and for the purposes set out in the procedure

I have read the information on the personal data processing policy

YES